Credit Analysis

Concept-First Notes + Exam Q&A + MCQ Bank + Mock Test — CA Intermediate

One complete file: concepts, worked Q&A, revision cheat-sheet, MCQs & a full mock test

Contents

What Credit Analysis Is & the Lender's Mindset

The Problem / Why this matters

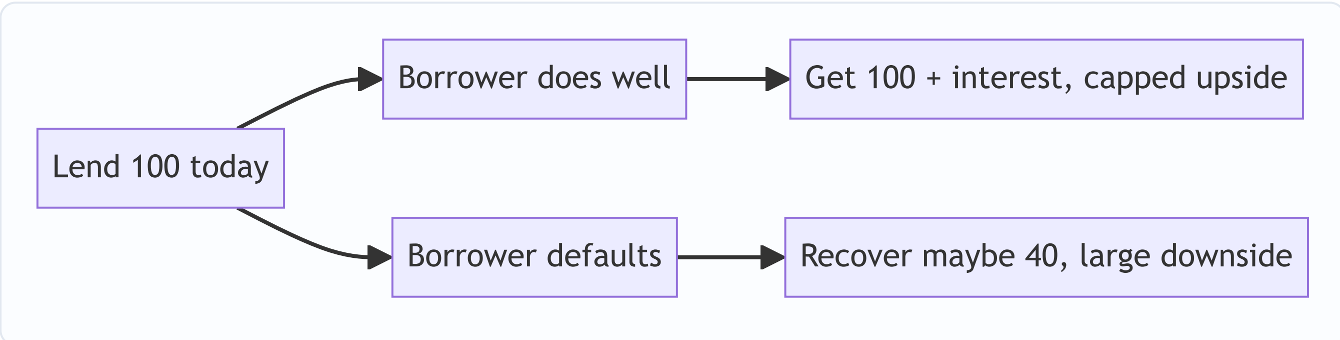
A lender hands over money today in exchange for a *promise* of repayment tomorrow. Unlike an equity investor, the lender does not share in the upside if the borrower doubles in size — the best case is simply getting paid back in full, on time, with interest. The whole discipline of credit analysis exists to answer one deceptively simple question: **will I be repaid, in full and on time, and what do I lose if I am not?** Get it wrong and a single default can wipe out the thin margin earned on dozens of good loans.

Core Idea

Credit analysis is the **structured assessment of a borrower's ability and willingness to service and repay debt**, plus the **loss the lender suffers if they don't**. It combines quantitative work (spreading financials, computing leverage and coverage ratios, modelling cash flow) with qualitative judgement (management quality, industry position, structure and covenants).

Why it works this way — the lender's asymmetry

The defining feature of credit is an **asymmetric payoff**. If a company you lent ₹100 to thrives, you still only get your ₹100 plus interest back. If it fails, you can lose most or all of it.



This asymmetry forces a **downside-first mindset**: the credit analyst spends most of their time asking "what could go wrong and how bad is it?" rather than "how big could this get?" It also explains why lenders obsess over **cash flow** (the source of repayment), **structure** (seniority, security, covenants that protect them) and **capital** (the equity cushion that absorbs losses before debt is hit).

Full technical content

Ability vs willingness to pay. Two independent questions: - *Ability* — does the business generate enough cash to service and repay debt? Assessed through cash flow, leverage and coverage. - *Willingness* — will they choose to pay even when stressed? Assessed through track record, governance and reputation (the "character" C).

A borrower can be able but unwilling (strategic default) or willing but unable (genuine distress). Both end in loss.

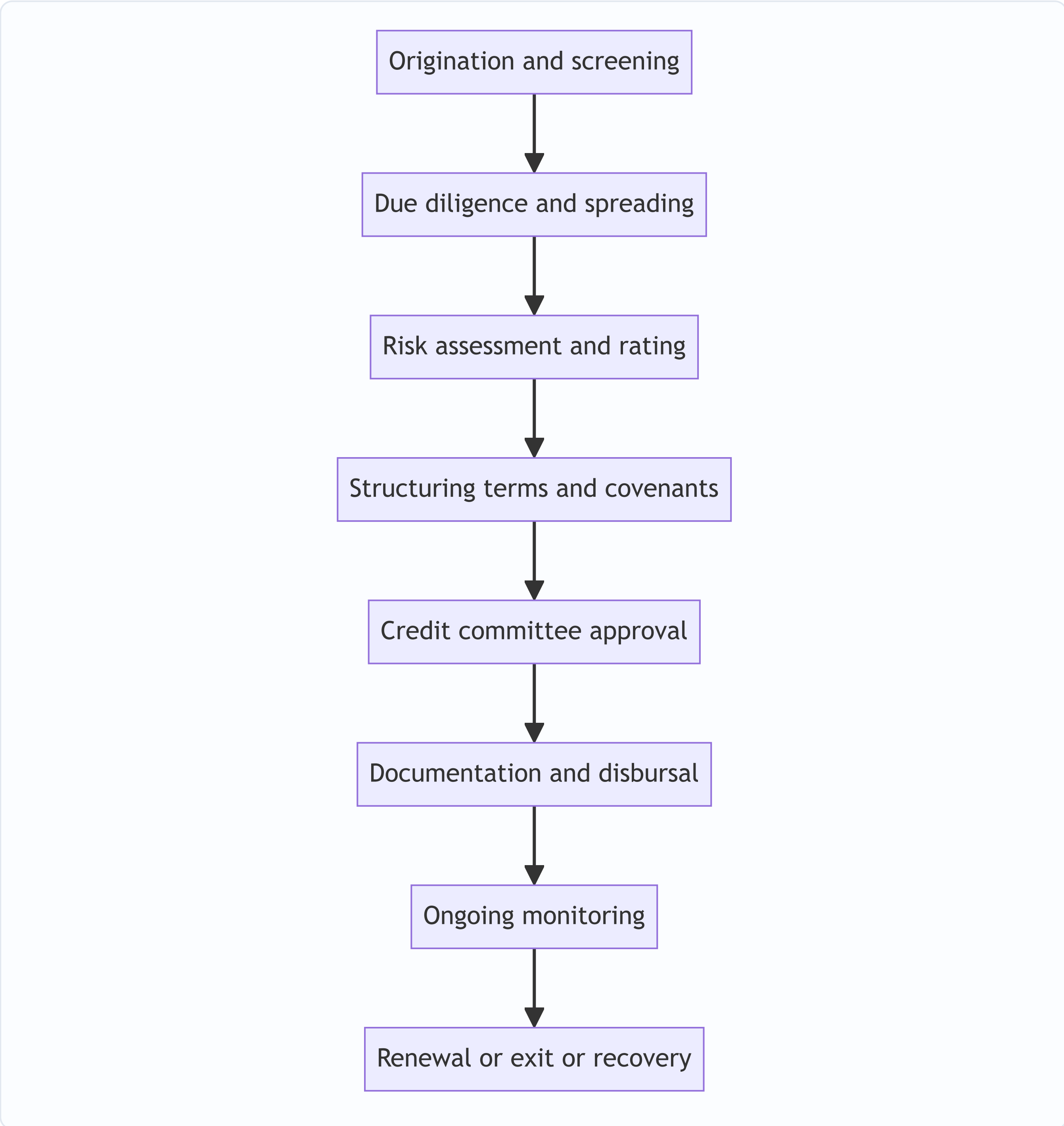
The two pillars of every credit view: | Pillar | Question | Tools | |---|---|---| | **Business risk** | How stable and defensible are the cash flows? | Industry analysis, competitive position, cyclicality, customer concentration | |

Financial risk | How much debt sits on those cash flows and can they service it? | Leverage (Debt/EBITDA), coverage (ICR, DSCR), liquidity |

The golden rule: **business risk sets the tolerable financial risk.** A stable utility can carry 6x Debt/EBITDA; a cyclical commodity trader cannot safely carry 3x.

Types of lending (the analysis flexes by type): - **Corporate / term lending** — funding capex or acquisitions, repaid from operating cash flow over years. - **Working-capital lending** — cash credit, overdraft, drawing-power based, repaid from the operating cycle. - **Project / infrastructure finance** — non-recourse, repaid only from the project's own ring-fenced cash flows. - **NBFC / financial-institution lending** — lending to a lender; assess their asset quality and ALM. - **Retail / SME** — statistical, scorecard-driven, portfolio approach.

The credit process (origination to exit):



Worked examples

Example 1 — Ability without willingness. Company A generates ₹500 cr of operating cash flow against ₹120 cr of debt service — comfortably able. But the promoter has a history of diverting funds to related parties and has defaulted on another group entity. *Credit view:* strong financials, but character risk caps the rating; require ring-fencing, escrow of cash flows, and tighter covenants, or decline.

Example 2 — Willingness without ability. Company B has an honest, committed management but sits in a collapsing sector with EBITDA of ₹40 cr against ₹75 cr of annual debt service (DSCR 0.53). No amount of good intent repays a ₹35 cr annual shortfall. *Credit view:* unable to service; restructuring or exit, not new lending.

Example 3 — Business risk framing financial risk. Two firms both at 4.0x Debt/EBITDA. Firm X is a regulated gas utility with contracted cash flows; Firm Y is a steel trader. Same leverage, very different risk: 4.0x is conservative for X and aggressive for Y. *Lesson:* leverage is only meaningful against the stability of the cash flow beneath it.

How it is tested in interviews

- **"What is credit analysis / what does a credit analyst do?"** — "Assess whether a borrower can and will repay its debt, and size the loss if it can't. I focus on cash flow as the source of repayment, leverage and coverage as the burden, and structure and covenants as my protection — always downside-first."
- **"How is credit different from equity analysis?"** — "Asymmetric payoff. Equity is paid for upside; credit is capped at par and punished on the downside, so I analyse the worst case, not the base case."
- **"What matters most in a credit — the business or the balance sheet?"** — "Both, but business risk sets how much financial risk is tolerable. Stable cash flows can support far more leverage than volatile ones."
- **"A company is profitable — is it a good credit?"** — "Not necessarily. Profit is accrual; debt is repaid with cash. I'd check operating cash flow, the maturity profile, and liquidity before answering."

Traps & common mistakes

- Confusing **profit with repayment capacity** — cash, not net income, services debt.
- Judging leverage **in isolation** from business risk.
- Ignoring **willingness** — great numbers under a promoter who has defaulted elsewhere is still a bad credit.
- Forgetting the lender's job is **loss avoidance**, not picking winners — you can be right on the company and still lose if the structure is weak.

First-principles recap

- Credit's payoff is asymmetric: capped upside, large downside — so analyse downside-first.
- Repayment needs both **ability** (cash flow) and **willingness** (character).
- **Business risk sets tolerable financial risk.**
- Cash flow repays debt; structure and covenants protect the lender; capital is the cushion.
- The output is a view on **probability of default** and **loss if it happens**.

Quick-reference

Concept	One-liner
Credit question	Repaid in full, on time — and loss if not?
Ability	Cash flow vs debt service
Willingness	Character, governance, track record
Business risk	Stability/defensibility of cash flows
Financial risk	Leverage + coverage on those flows
Golden rule	Business risk sets tolerable financial risk
Source of repayment	Operating cash flow (not profit)

The 5 Cs of Credit

The Problem / Why this matters

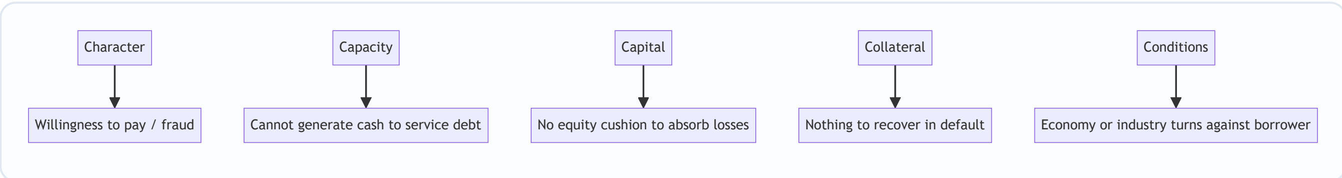
Before spreadsheets and rating models, lenders needed a **checklist that would not let them forget anything that has historically killed a loan**. The 5 Cs — Character, Capacity, Capital, Collateral, Conditions — are that checklist. They are the oldest framework in credit and still the first thing an interviewer expects you to reel off, because they organise everything else you do into five buckets that map to the real ways loans go bad.

Core Idea

A sound credit decision requires all five: an honest borrower (**Character**) with enough cash flow (**Capacity**), a real equity cushion (**Capital**), backed by assets (**Collateral**), operating in a supportive environment (**Conditions**). Weakness in one can sometimes be offset by strength in another — strong collateral can rescue thin capacity — but a fatal flaw in Character usually cannot be offset at all.

Why it works this way

Each C corresponds to a distinct failure mode observed over centuries of lending:



Miss any one and you have re-created a historical loss. The 5 Cs force breadth so that a strong number in one area doesn't blind you to a hole in another.

Full technical content

1. Character — the willingness to pay. The borrower's integrity, track record and credit history. Evidence: past repayment behaviour, credit-bureau records (CIBIL), conduct of existing accounts, related-party dealings, promoter reputation, quality and stability of management, auditor changes and qualifications. Character is the one C that, if failed, is rarely curable — you cannot structure your way around a borrower who won't pay.

2. Capacity — the ability to pay. The cash-generating ability to service and repay debt. This is the quantitative heart: operating cash flow, EBITDA, free cash flow, and the coverage ratios (interest coverage, DSCR). Capacity answers "does the business throw off enough cash to cover the debt, with headroom?"

3. Capital — the owner's stake. The equity the owners have invested and the net worth cushion that absorbs losses before creditors are touched. High capital (low leverage) means the owners lose first and have "skin in the game," aligning incentives. Measured by Debt/Equity, gearing, tangible net worth.

4. Collateral — the fallback. Assets pledged as security that the lender can seize and sell if the borrower defaults. Reduces **loss given default**, not probability of default. Quality matters: liquid, easily valued, unencumbered collateral (property, plant, receivables, cash) beats specialised or fast-depreciating assets. Assessed on realisable (not book) value with a haircut.

5. Conditions — the environment. The macro and industry backdrop, the purpose of the loan, and the terms. Includes interest-rate cycle, sector outlook, regulation, competitive dynamics, and how the funds will be used. A perfectly good borrower can be sunk by a sector collapse.

C	Failure mode it guards against	Key evidence
Character	Won't pay / fraud	Bureau record, track record, governance
Capacity	Can't generate cash	Cash flow, EBITDA, ICR, DSCR
Capital	No loss-absorbing cushion	D/E, gearing, net worth
Collateral	Nothing to recover	Security value, haircut, seniority
Conditions	Environment turns	Industry, macro, loan purpose

Worked examples

Example 1 — a hole in one C. A borrower scores well on Capacity (DSCR 2.0x), Capital (D/E 0.5), Collateral (property worth 1.5x the loan) and Conditions (growing sector) — but the promoter defaulted on a sister company last year (Character). *Decision:* the Character flaw dominates; either decline or lend only with ring-fenced cash-flow escrow and personal guarantees, at a punitive spread.

Example 2 — Collateral offsetting Capacity. A property developer has lumpy, uncertain cash flow (weak Capacity) but pledges completed, saleable inventory worth 2x the loan (strong Collateral). *Decision:* lendable as an asset-backed facility with conservative loan-to-value and drawdown tied to sales — Collateral compensates for Capacity.

Example 3 — Conditions sinking a good borrower. A textile exporter with clean Character, solid Capacity and Capital faces a sudden 30% currency appreciation and a new import tariff in its main market (Conditions). Historical strength is irrelevant to a demand shock. *Decision:* stress the cash flows for the new environment before extending credit.

How it is tested in interviews

- **"What are the 5 Cs of credit?"** — Name them and give one line each; then add the insight that **Character is the one you can't structure around**, and **Capacity is where the numbers live**.
- **"Which C is most important?"** — "Character for willingness and Capacity for ability — a loan needs both. Collateral and Capital are cushions that reduce loss, not substitutes for cash flow."
- **"How do Collateral and Capital differ?"** — "Capital is the owners' equity absorbing losses first; Collateral is specific assets I can seize. Capital reduces the chance of default; Collateral reduces my loss if it happens."
- **"How would you assess Character?"** — "Bureau/CIBIL record, conduct of existing accounts, related-party transactions, promoter and management track record, and any auditor red flags."

Traps & common mistakes

- Treating Collateral as a substitute for cash flow — **"lend against cash flow, not collateral."** Collateral is plan B, not plan A.
- Skipping Character because the numbers look good.
- Confusing Capital (equity cushion, reduces PD) with Collateral (assets, reduces LGD).
- Ignoring Conditions / loan purpose — a good borrower funding a bad project is still a bad loan.

First-principles recap

- Five buckets, each guarding a historical way loans fail.
- **Character** (willingness) and **Capacity** (ability) are the core; a loan needs both.
- **Capital** reduces probability of default; **Collateral** reduces loss given default.
- **Conditions** can sink even a strong borrower.
- A fatal flaw in one C usually can't be fully offset by strength in another — especially Character.

Quick-reference

C	One-liner	Maps to
Character	Will they pay?	PD / willingness
Capacity	Can they pay?	PD / cash flow
Capital	Owners' cushion	PD / leverage
Collateral	Fallback assets	LGD / recovery
Conditions	Environment & purpose	Systematic risk

Financial Spreading & Normalizing Statements

The Problem / Why this matters

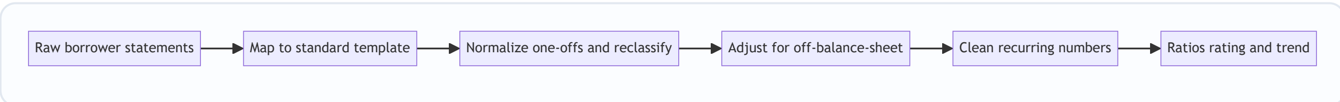
Every borrower presents its financials differently — different line items, different groupings, aggressive classifications, one-off gains dressed up as operating profit. Before you can compare a borrower to peers, to covenants, or to last year, you must **restate its financials into a single, standard, apples-to-apples template**. That process is called *spreading*, and it is the unglamorous foundation on which every ratio, rating and decision rests. Spread badly and every number downstream is wrong.

Core Idea

Spreading = taking the borrower's raw income statement, balance sheet and cash flow and re-mapping every line into the lender's standardised format, **normalizing** for distortions (one-offs, related-party items, accounting choices, off-balance-sheet debt) so the resulting numbers reflect the *true, recurring, economic* picture.

Why it works this way

Ratios are only comparable if the inputs are defined consistently. If Firm A puts forex loss in "other income" and Firm B puts it in "operating expense," their operating margins aren't comparable until you standardise. And reported EBITDA can be inflated by non-recurring gains or hidden leverage (operating leases, factored receivables) that a lender must pull back onto the balance sheet.



Full technical content

What gets spread. Three statements, usually 3–5 years plus interim, into a fixed template: - **Income statement:** revenue, COGS, gross profit, operating expenses, EBITDA, D&A, EBIT, interest, non-operating items, PBT, tax, PAT — with recurring vs non-recurring separated. - **Balance sheet:** grouped into a lender view — current assets (cash, receivables, inventory), non-current assets, current liabilities (payables, short-term debt), long-term debt, net worth. - **Cash flow:** CFO, CFI, CFF, and derived free cash flow / cash available for debt service.

Normalizations the analyst makes: | Adjustment | Why | |---|---| | Strip **one-off gains/losses** (asset sales, forex, litigation) | Isolate recurring earning power | | Reclassify **misplaced items** (forex in "other income") | Make margins comparable | | Add back **operating leases** as debt (IFRS 16 style) | Capture hidden leverage | | Add back **factored/discounted receivables** | Restore true debt and receivables | | Adjust **related-party** revenue/purchases | Remove non-arm's-length distortion | | Normalize **owner compensation** (private firms) | Reflect market-rate cost | | Treat **contingent liabilities/guarantees** | Capture off-balance-sheet risk | | Remove **capitalized expenses** that should be expensed | Undo profit inflation |

Quality checks while spreading: does the balance sheet balance? Does the change in retained earnings tie to net income minus dividends? Does the cash flow reconcile to the change in the cash line? Inconsistencies are red flags, not rounding.

Output: a clean multi-year trend of standardized figures ready for leverage, coverage, liquidity and profitability ratios, and for comparison against covenants and peers.

Worked examples

Example 1 — normalizing EBITDA. Reported EBITDA is ₹120 cr. It includes a ₹25 cr one-off gain on sale of land and ₹10 cr of forex gain (non-operating), but excludes ₹15 cr of operating-lease rent that under IFRS 16 is really debt-servicing. *Normalized EBITDA* = 120 − 25 − 10 + 15 = **₹100 cr**. Leverage on ₹400 cr debt is 4.0x on the clean number vs a flattering 3.3x on the reported one. The lender uses 4.0x.

Example 2 — restoring hidden debt. A firm factors ₹80 cr of receivables (sold to a financier, off balance sheet) and has ₹120 cr of operating-lease commitments (~₹60 cr capitalized). Reported debt ₹300 cr. *Adjusted debt* = 300 + 80 + 60 = **₹440 cr** — nearly 50% higher. Debt/EBITDA jumps from 3.0x to 4.4x. This single adjustment can change the rating.

Example 3 — related-party revenue. ₹200 cr of a firm's ₹500 cr revenue is to a promoter-owned entity at above-market prices. Stripping the ₹30 cr of inflated margin lowers normalized EBITDA and reveals the standalone business is weaker than it looks.

How it is tested in interviews

- **"What is financial spreading?"** — "Restating a borrower's statements into a standard template and normalizing for one-offs, reclassifications and off-balance-sheet items, so ratios are clean and comparable."
- **"What would you adjust for when spreading?"** — Name: one-off gains/losses, operating leases, factored receivables, related-party items, contingent liabilities, capitalized costs.
- **"Reported EBITDA is 120 with a 25 one-off gain and 15 of lease rent — what's your number?"** — Walk the 120 − 25 + 15 = 110 logic (adjust for the specific items given) and explain why the clean number drives leverage.
- **"Why add operating leases to debt?"** — "They're a fixed, debt-like obligation; ignoring them understates leverage and overstates coverage."

Traps & common mistakes

- Taking **reported EBITDA** at face value.
- Missing **off-balance-sheet debt** (leases, factoring, guarantees) — the most common way leverage is understated.
- Not separating **recurring from one-off** — a firm can look profitable on the back of asset sales.
- Forgetting to **cross-check** the three statements tie out; distortions often reveal manipulation.

First-principles recap

- Spreading standardizes messy statements so numbers are comparable.
- Normalize for **one-offs, reclassifications, and off-balance-sheet debt**.
- Restore hidden leverage (leases, factoring) — it changes the rating.
- Always verify the statements tie out; breaks are red flags.

- Clean, recurring numbers in — reliable ratios and ratings out.

Quick-reference

Step	Action
Map	Raw lines into standard template
Strip	One-off gains/losses
Reclassify	Items in the wrong bucket
Restore	Leases, factored receivables, guarantees to debt
Check	Balance sheet balances; CF ties to cash
Output	Clean multi-year recurring numbers

Leverage & Coverage Ratios

The Problem / Why this matters

Once the financials are spread, the lender needs a small set of numbers that answer, at a glance: **how much debt is this, relative to the cash flow and equity beneath it, and can that cash flow comfortably cover the payments?** Leverage and coverage ratios are those numbers. They are the language of every credit committee, every covenant, and every rating methodology — and interviewers expect you to define them precisely and know good-vs-bad ranges.

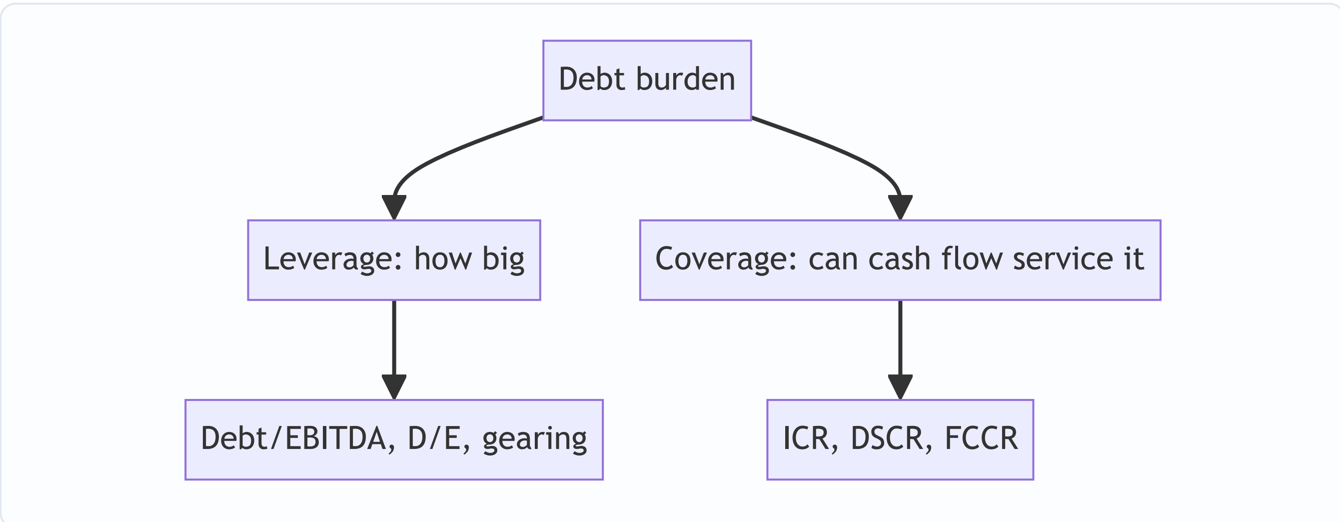
Core Idea

- **Leverage ratios** measure the *size* of the debt burden relative to earnings or equity (how deep the hole).
- **Coverage ratios** measure the *serviceability* — whether recurring cash flow comfortably covers interest and principal (can they climb out).

You need both: a firm can have modest leverage but thin coverage if rates spike, or heavy leverage but strong coverage if cash flow is huge and stable.

Why it works this way

Leverage tells you the stock of the problem; coverage tells you the flow. Debt is repaid from cash flow, so coverage is often the more binding constraint, while leverage sets the medium-term risk. Rating agencies and covenants use both because each catches a different failure: over-borrowing (leverage) and cash-flow shortfall (coverage).



Full technical content

Leverage ratios | Ratio | Formula | Reads | |---|---|---| | Debt / EBITDA | Total debt ÷ EBITDA | Years of earnings to repay debt | | Net Debt / EBITDA | (Debt − cash) ÷ EBITDA | Same, netting surplus cash | | Debt / Equity (gearing) | Total debt ÷ tangible net worth | Debt relative to owners' cushion | | Debt / Capital | Debt ÷ (Debt + Equity) | Share of capital from debt |

Rules of thumb (adjust for sector): Debt/EBITDA below ~2x is conservative, 2–4x moderate, 4–6x aggressive, above 6x highly leveraged; investment-grade corporates typically sit under ~3–3.5x. Utilities tolerate more; cyclicals less.

Coverage ratios | Ratio | Formula | Reads | |---|---|---| | Interest Coverage (ICR) | EBIT ÷ Interest | Times operating profit covers interest | | EBITDA Interest Coverage | EBITDA ÷ Interest | Cash-based interest cover | | **DSCR** | Cash available for debt service ÷ (Interest + Principal) | Can it cover interest AND principal | | FCCR | (EBITDA − capex − tax) ÷ (Interest + Principal + leases) | Broadest fixed-charge cover |

DSCR is the most important for term loans and project finance. **DSCR > 1.0** means the borrower generates enough to meet its full debt service; lenders usually require a cushion (e.g., 1.2x–1.5x). Below 1.0x the borrower must draw on reserves or refinance.

How the two interact. A firm at 5x Debt/EBITDA (high leverage) with EBITDA/interest of 6x and DSCR of 1.6x may still be fine short-term — cash flow covers payments — but is fragile to an earnings dip. A firm at 3x leverage but DSCR of 1.05x is closer to the edge on serviceability. Read them together.

Worked examples

Example 1 — full ratio set. EBITDA ₹100 cr, D&A ₹20 cr (so EBIT ₹80 cr), interest ₹25 cr, principal due ₹35 cr, capex ₹15 cr, tax ₹10 cr, total debt ₹400 cr, cash ₹40 cr, tangible net worth ₹200 cr. - Debt/EBITDA = 400/100 = **4.0x**; Net Debt/EBITDA = 360/100 = **3.6x** - D/E = 400/200 = **2.0x** - ICR (EBIT) = 80/25 = **3.2x**; EBITDA interest cover = 100/25 = **4.0x** - DSCR = (EBITDA − tax) / (interest + principal) = (100 − 10)/(25 + 35) = 90/60 = **1.5x** - FCCR = (100 − 15 − 10)/(25 + 35) = 75/60 = **1.25x** *View:* moderately-high leverage (4.0x) but healthy DSCR (1.5x) and FCCR (1.25x) — serviceable, with some cushion. Watch leverage on any EBITDA decline.

Example 2 — coverage binding before leverage. Firm at only 2.5x Debt/EBITDA but EBITDA barely covers interest plus a heavy amortization schedule: DSCR 1.02x. Low leverage is deceptive — a small earnings dip breaches DSCR. *Restructure the amortization or reject.*

Example 3 — leverage binding. Firm at 6.5x Debt/EBITDA with DSCR 1.4x today (cheap current rates). Serviceable now, but a refinancing at higher rates or an EBITDA fall makes 6.5x untenable. *High leverage = refinancing risk even with adequate current coverage.*

How it is tested in interviews

- **"What's the difference between leverage and coverage?"** — "Leverage sizes the debt burden (Debt/EBITDA, D/E); coverage tests whether cash flow services it (ICR, DSCR). I need both."
- **"What is DSCR and why does it matter?"** — "Cash available for debt service over interest plus principal. Above 1.0 means the borrower can meet full debt service; lenders want a 1.2–1.5x cushion. It's the key serviceability metric for term loans."
- **"Company at 3x Debt/EBITDA — safe?"** — "Depends on the sector's stability and on coverage. 3x is fine for a utility, aggressive for a cyclical; and I'd check DSCR before concluding."
- **"Walk me through the ratios you'd compute."** — List leverage (Debt/EBITDA, D/E) then coverage (ICR, DSCR, FCCR), and note DSCR is usually the binding covenant.

Traps & common mistakes

- Using **EBIT vs EBITDA** inconsistently in coverage — state which.
- Forgetting DSCR includes **principal**, not just interest (that's the whole point vs ICR).
- Reading leverage **without the sector** context.
- Ignoring **off-balance-sheet debt** (from spreading) — it inflates the true ratios.
- Treating a single year's ratio as the truth — use the **trend** and a stressed case.

First-principles recap

- Leverage = size of debt (Debt/EBITDA, D/E); coverage = serviceability (ICR, DSCR, FCCR).
- **DSCR** is the key term-loan metric; > 1.0 with a cushion is the target.
- ICR covers only interest; DSCR adds principal; FCCR adds capex/leases.
- Interpret every ratio against **sector stability** and the **trend**, not in isolation.
- Use spread, off-balance-sheet-adjusted numbers.

Quick-reference

Ratio	Formula	Good-ish
Debt/EBITDA	Debt/EBITDA	< 3–3.5x (IG)
Net Debt/EBITDA	(Debt−Cash)/EBITDA	lower is better
D/E	Debt/Net worth	< 1–2x sector-dependent
ICR	EBIT/Interest	> 3x comfortable
DSCR	CADS/(Int+Principal)	> 1.2–1.5x
FCCR	(EBITDA−capex−tax)/fixed charges	> 1.2x

Cash Flow Analysis for Credit

The Problem / Why this matters

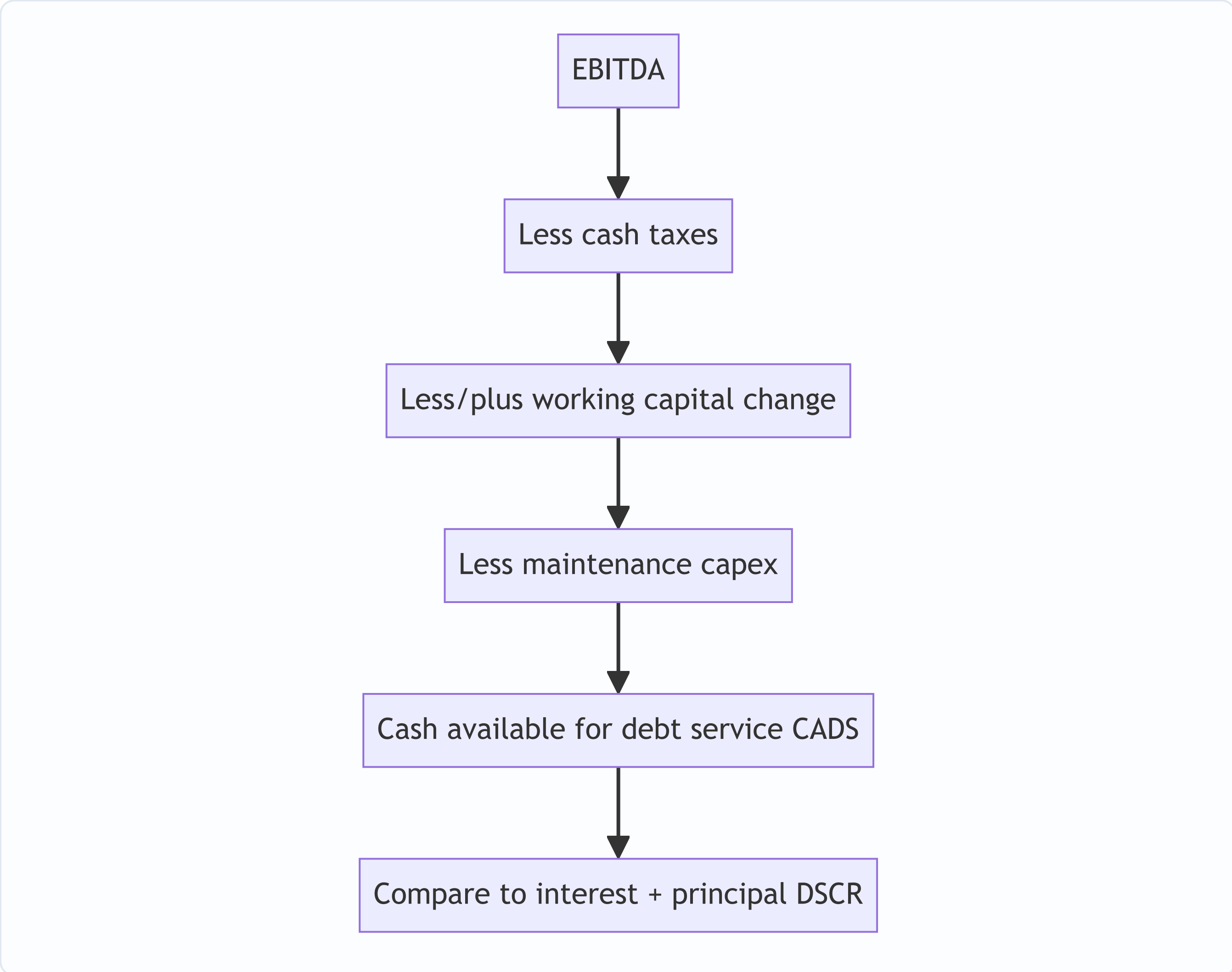
Debt is not repaid with profit, revenue, or EBITDA — it is repaid with **cash**. A company can report record net income and still default because the "profit" is tied up in receivables and inventory, or consumed by capex and taxes. The single most important skill in credit is tracing where cash actually comes from and whether enough is genuinely **available to service debt** after everything else the business must pay. This is where good credit analysts separate themselves from ratio-readers.

Core Idea

Cash-flow analysis for credit rebuilds the borrower's cash from operations, strips out what the business unavoidably consumes (working-capital swings, maintenance capex, taxes), and arrives at **cash available for debt service (CADS)** — the real number that repays lenders. From there you test it against required debt service and stress it.

Why it works this way

EBITDA is a starting proxy for cash, but it ignores three real cash costs: **taxes**, **working-capital changes**, and **capex**. A growing firm ploughs cash into receivables and inventory; a capital-intensive firm ploughs it into equipment. Both can have strong EBITDA and negative free cash flow. Lenders therefore walk EBITDA down to the cash that actually reaches the debt-service line.



Full technical content

The cash-flow waterfall (from EBITDA to CADS): | Step | Item | |---|---| | Start | EBITDA (normalized, recurring) | | − | Cash taxes | | ± | Change in working capital (increase = use of cash) | | = | Operating cash flow (CFO) | | − | Maintenance capex (the capex needed to keep running) | | = | **Cash available for debt service (CADS)** | | − | Interest + scheduled principal | | = | Cash after debt service (build/deplete reserves) | | − | Growth capex, dividends | (discretionary — can be cut in stress) |

Key distinctions: - **Recurring vs one-off cash** — cash from asset sales or a working-capital release is not repeatable; don't rely on it for term debt. - **Maintenance vs growth capex** — only maintenance capex is unavoidable; growth capex can be deferred in stress, so it sits below the debt-service line. - **Quality of earnings** — compare **CFO to net income**: if profit rises while CFO falls, earnings quality is poor (receivables/inventory build or aggressive accruals) — a classic credit red flag. - **Free cash flow (FCF)** = CFO − total capex; sustained positive FCF is the deleveraging engine.

Sources of repayment hierarchy (what actually repays the loan): 1. **Primary** — operating cash flow (what you underwrite to). 2. **Secondary** — refinancing / asset sales (fragile, market-dependent). 3. **Tertiary** — collateral enforcement (last resort, lossy). A sound credit is repaid from the primary source; relying on secondary/tertiary is a weak credit.

Worked examples

Example 1 — EBITDA to CADS. EBITDA ₹100 cr; cash tax ₹15 cr; working capital increased ₹20 cr (use of cash); maintenance capex ₹18 cr. CADS = 100 − 15 − 20 − 18 = **₹47 cr**. Against interest ₹20 cr + principal ₹15 cr = ₹35 cr debt service, DSCR = 47/35 = **1.34x**. Note: the ₹100 EBITDA looked like huge cover, but real CADS is less than half of it.

Example 2 — profit up, cash down. Net income grows 20% to ₹60 cr, but receivables jumped ₹90 cr on aggressive credit terms, so CFO fell to ₹5 cr. *Red flag:* the growth is being financed by extending credit to customers, not by cash generation — leverage will rise and liquidity tighten. A ratio-only analyst misses this; a cash-flow analyst catches it.

Example 3 — capex hiding the risk. Two firms both at EBITDA ₹80 cr and 3x leverage. Firm A is asset-light (maintenance capex ₹5 cr → CADS ₹65 cr); Firm B is capital-intensive (maintenance capex ₹45 cr → CADS ₹25 cr). Same EBITDA and leverage, very different debt-service capacity. *Lesson:* EBITDA-based leverage flatters capital-intensive borrowers.

How it is tested in interviews

- **"Why do you focus on cash flow, not profit?"** — "Debt is repaid with cash. Profit is accrual and can be tied up in working capital or consumed by capex and tax. I walk EBITDA down to cash available for debt service."
- **"Walk me from EBITDA to cash available for debt service."** — EBITDA − cash tax ± working-capital change − maintenance capex = CADS; then compare to interest + principal.
- **"Net income is up but operating cash flow is down — concern?"** — "Yes — a classic earnings-quality red flag. Likely a receivables or inventory build, or aggressive accruals. It means the growth isn't cash-generative and leverage/liquidity will worsen."
- **"What repays a loan?"** — "Primarily operating cash flow; refinancing and asset sales are secondary; collateral is the last resort. I underwrite to the primary source."

Traps & common mistakes

- Equating **EBITDA with cash** — it ignores tax, working capital and capex.
- Counting **one-off** cash (asset sales, WC release) as repeatable.
- Not splitting **maintenance vs growth capex** — overstating unavoidable spend or ignoring it.
- Missing the **CFO-vs-net-income** quality signal.
- Relying on **secondary/tertiary** repayment sources for a term loan.

First-principles recap

- Cash, not profit, repays debt — walk EBITDA down to CADS.
- CADS = EBITDA − cash tax ± Δ working capital − maintenance capex.
- Compare CFO to net income for earnings quality.
- Underwrite to the **primary** source (operating cash flow); collateral is last resort.
- Only maintenance capex is unavoidable; growth capex and dividends can be cut in stress.

Quick-reference

Item	Note
CADS	EBITDA − cash tax ± ΔWC − maint. capex
DSCR	CADS ÷ (Interest + Principal)
FCF	CFO − total capex
Quality flag	CFO falling while NI rising
Repayment order	Operating CF > refinancing/asset sales > collateral

Business & Industry Risk Assessment

The Problem / Why this matters

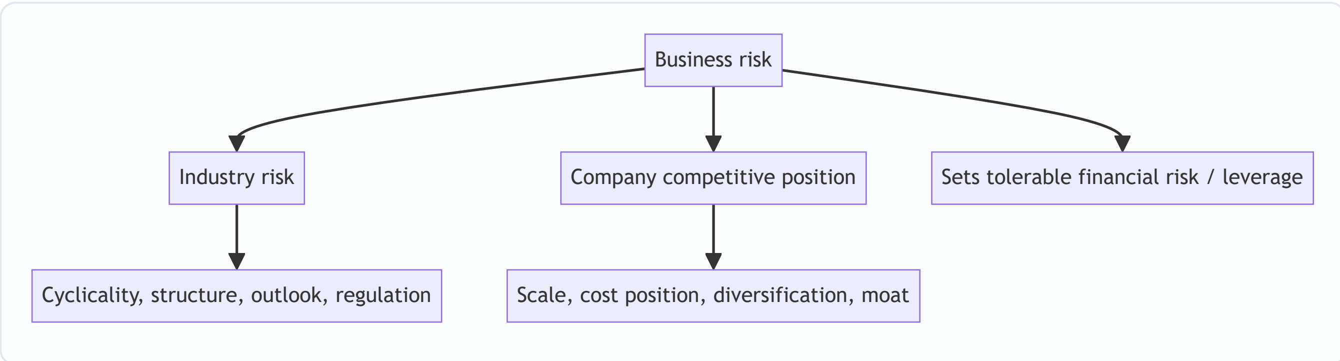
Two companies can have identical financials today and completely different creditworthiness, because one operates in a stable, defensible industry and the other in a volatile, competitive one. **Business risk — the stability and defensibility of a borrower's cash flows — determines how much financial risk (leverage) is safe.** Ignore it and you will lend prudently-leveraged money to a business whose cash flows can halve in a downturn. This qualitative work is what turns a spreadsheet into a credit judgement.

Core Idea

Business-risk analysis asks: **how reliable are the future cash flows that will repay this debt?** It assesses the industry (cyclicality, structure, outlook) and the company's position within it (scale, cost position, diversification, competitive moat), to set the tolerable leverage and the right rating.

Why it works this way

Debt service is fixed; cash flow is variable. The more variable and less defensible the cash flow, the more likely it falls below the fixed debt service in a bad year. So a lender pairs **low leverage with high business risk** and can accept **high leverage with low business risk**.

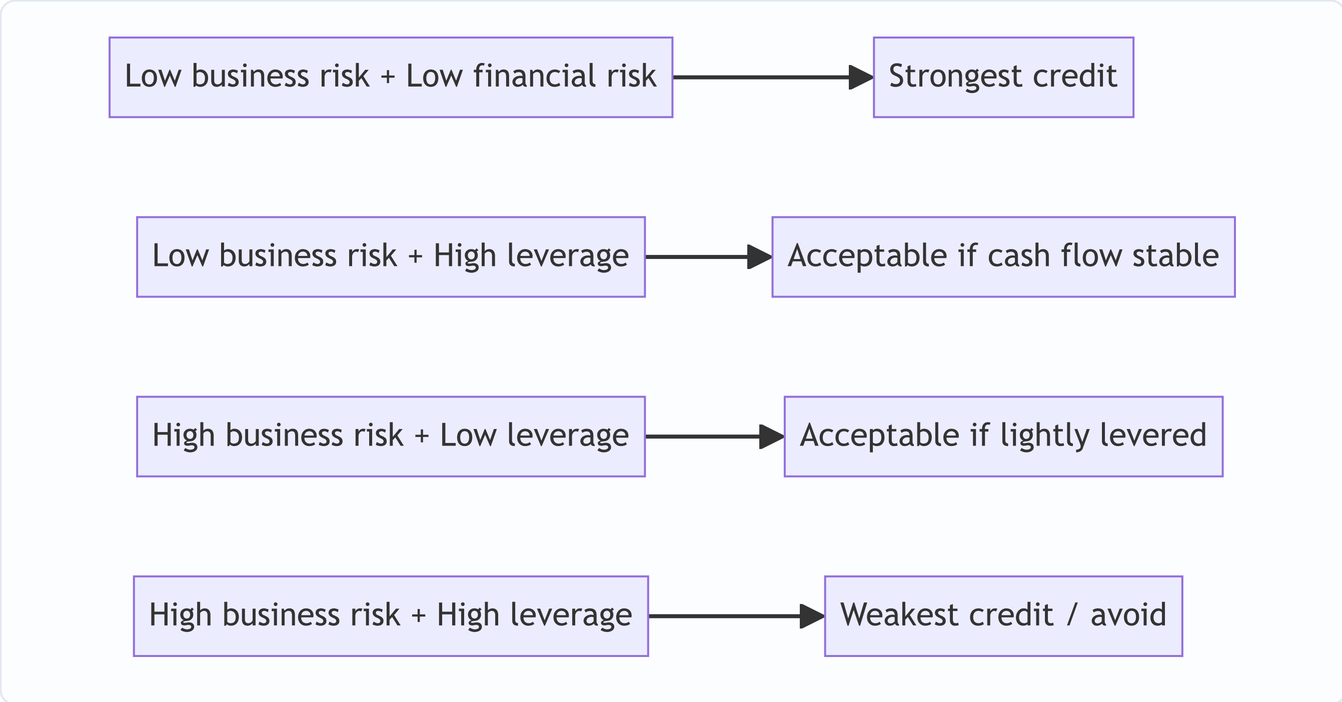


Full technical content

Industry risk factors: | Factor | What to assess | |---|---| | Cyclicality | How much do revenues/margins swing with the economy? | | Industry structure | Fragmented vs consolidated; a Porter five-forces read | | Growth & maturity | Growing, mature, or declining demand | | Capital intensity | High fixed costs magnify downturns (operating leverage) | | Regulation | Tariffs, price controls, licensing, ESG rules | | Barriers to entry | Do incumbents earn durable returns? | | Technology / disruption | Risk of obsolescence |

Company competitive-position factors: | Factor | What to assess | |---|---| | Scale & market share | Cost and pricing advantages of size | | Cost position | Low-cost producer survives downturns | | Diversification | By product, customer, geography, supplier | | Customer/supplier concentration | Reliance on a few counterparties | | Moat | Brand, switching costs, network, IP, contracts | | Management quality | Strategy, execution, capital discipline | | Contract/recurring revenue | Visibility and stickiness of cash flow |

The synthesis — the risk matrix. Combine business risk and financial risk into a rating grid:



Rating agencies formalize exactly this: a business-risk score and a financial-risk score map to an anchor rating.

Worked examples

Example 1 — same leverage, different risk. A regulated electricity distributor and a steel producer both at 4.0x Debt/EBITDA. The utility has contracted, inflation-linked, recession-resistant cash flows (low business risk) — 4.0x is comfortable, investment-grade. The steel producer's EBITDA can halve in a down-cycle (high business risk) — 4.0x could become 8.0x in a trough, sub-investment-grade. *Same number, opposite conclusions.*

Example 2 — concentration risk. A component maker earns 70% of revenue from a single auto OEM. Financials look solid at 2.5x leverage, but the loss of that one customer would be catastrophic. *Business risk (concentration) caps the rating despite low leverage; require diversification covenants or a lower exposure.*

Example 3 — cost position in a downturn. In a commodity glut, prices fall to the industry's marginal cost. The low-cost producer (first-quartile cost) stays cash-positive; the high-cost producer bleeds. Same industry, same leverage — the low-cost producer is a far better credit because it survives the trough.

How it is tested in interviews

- **"How does business risk affect a credit decision?"** — "It sets the tolerable leverage. Stable, defensible cash flows can support more debt; volatile ones can't. I pair low leverage with high business risk."
- **"Two firms at 4x leverage — how do you tell them apart?"** — Use the utility-vs-cyclical example: same leverage, different cash-flow stability, so different ratings.
- **"What business-risk factors do you look at?"** — Industry cyclicality/structure/regulation and company scale/cost-position/diversification/moat/management.
- **"Why does customer concentration matter for credit?"** — "It creates a single point of failure — losing one counterparty can wipe out cash flow regardless of current leverage."

Traps & common mistakes

- Judging a credit on **financials alone**, ignoring cash-flow stability.
- Treating all sectors at a given leverage as equal.
- Missing **concentration** (customer, supplier, geography, product).
- Overrating a firm's position without checking its **cost quartile** — position in a downturn is what matters.
- Ignoring **regulatory/technology disruption** that can reset an entire industry.

First-principles recap

- Business risk = stability and defensibility of future cash flows.
- It **sets the tolerable financial risk** (leverage).
- Assess industry (cyclicality, structure, regulation) and company (scale, cost, diversification, moat, management).
- Concentration and cost position are decisive in stress.
- Combine business and financial risk into the rating.

Quick-reference

Dimension	Key questions
Industry	Cyclical? Consolidated? Regulated? Growing?
Position	Scale? Low-cost? Diversified? Moat?
Concentration	One customer/supplier/geography?
Synthesis	Low business risk → tolerate more leverage
Rating	Business-risk score × financial-risk score → anchor

Debt Capacity & Structuring

The Problem / Why this matters

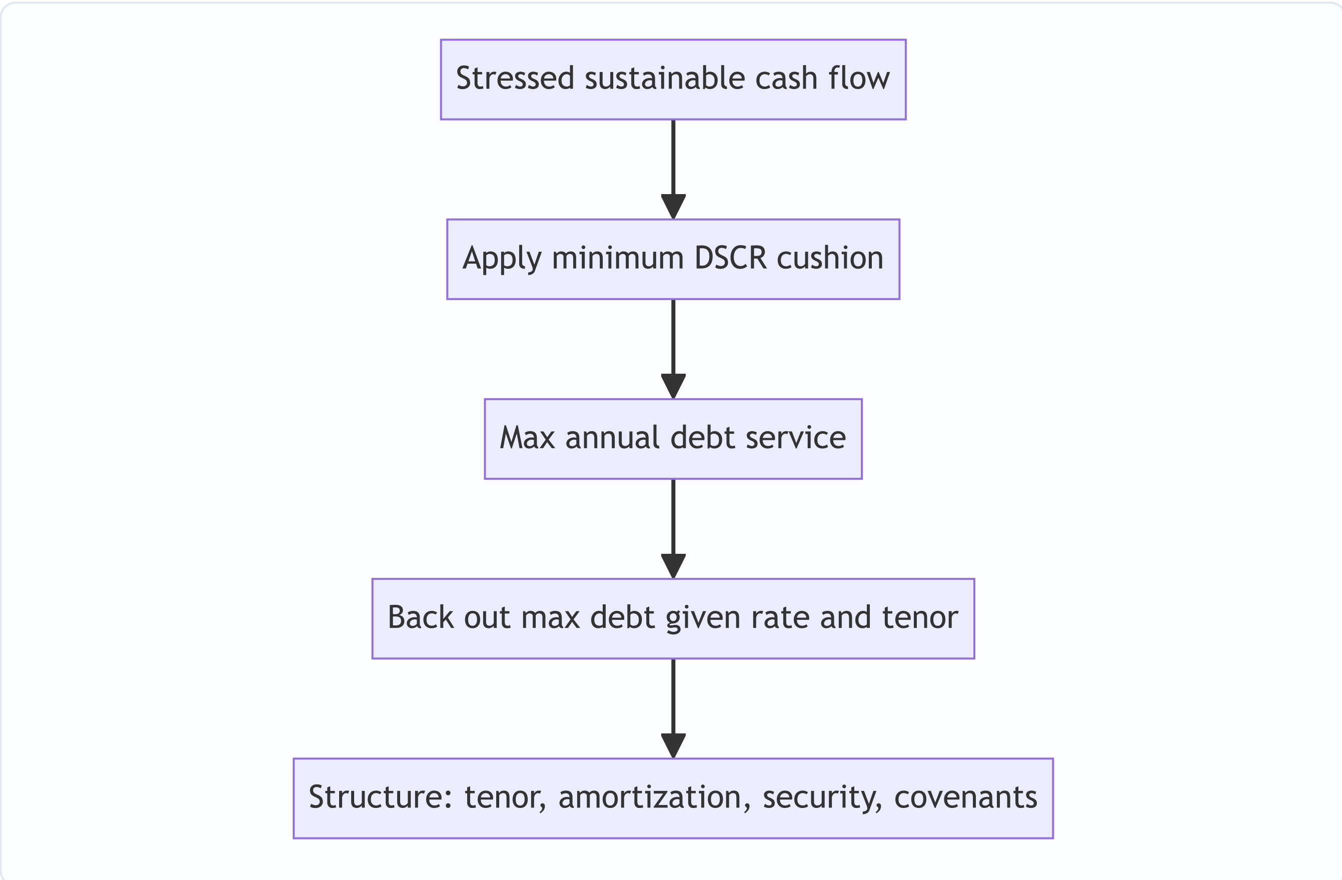
Approving a borrower is only half the job. The other half is deciding **how much to lend, on what terms, repaid how and over how long, and with what protections** — so that the loan is serviceable even if things go moderately wrong. A good borrower financed with the wrong structure (too much debt, too fast an amortization, a bullet at a bad time) still defaults. Structuring is where the analyst turns a credit view into a specific, safe transaction.

Core Idea

Debt capacity is the maximum debt a borrower can support from its *stressed* cash flow while keeping coverage above a safe threshold. **Structuring** then shapes the facility — size, tenor, amortization, security, pricing, covenants — to match the borrower's cash-flow profile and protect the lender.

Why it works this way

Debt service must fit inside cash flow with a cushion, in bad years as well as good. So capacity is sized off a *stressed* case, not the base case, and the repayment schedule is shaped to the cash the business actually throws off over time. Matching structure to cash flow is what keeps DSCR above 1.0 through the cycle.



Full technical content

Sizing debt capacity. Start from **sustainable, stressed** EBITDA/cash flow, apply a required minimum DSCR, and solve for the debt the cash flow can service. - Max annual debt service = CADS ÷ required DSCR. - Given an interest rate and tenor, back out the maximum principal an annuity of that debt service supports. - Cross-check against a maximum leverage (e.g., Debt/EBITDA cap) appropriate to the sector.

Structuring levers: | Lever | Choice | Effect | |---|---|---| | **Amount** | Sized to capacity | Keeps DSCR safe | | **Tenor** | Match asset/cash-flow life | Long assets → long tenor | | **Amortization** | Bullet, straight-line, sculpted, balloon | Shape repayment to cash flow | | **Seniority** | Senior secured → subordinated | Drives recovery/pricing | | **Security** | Fixed/floating charge, pledge, guarantee | Reduces LGD | | **Pricing** | Spread over benchmark | Compensates for risk | | **Covenants** | Maintenance + incurrence | Early-warning + control |

Repayment shapes: - **Bullet** — interest only, principal at maturity; maximizes refinancing risk. - **Straight-line amortization** — equal principal instalments; de-risks steadily. - **Sculpted** — instalments track projected cash flow (common in project finance to hold DSCR near constant). - **Balloon** — partial amortization plus a large final payment.

Match structure to cash-flow profile: stable, growing cash flow can take a bullet or back-loaded schedule; lumpy or declining cash flow needs faster front-loaded amortization while the cash is there. Tenor should not exceed the economic life of what's being financed ("don't fund a 3-year asset with 10-year debt, or a 20-year toll road with a 5-year bullet").

Worked examples

Example 1 — sizing capacity. Stressed CADS ₹60 cr; required minimum DSCR 1.5x. Max annual debt service = 60/1.5 = **₹40 cr**. If interest is ~10% and the lender wants a 5-year amortizing loan, the ₹40 cr/year annuity supports roughly **₹150 cr** of debt (annuity PV at 10%, 5 yrs ≈ 3.79 × 40 ≈ 151). Cross-check leverage: if EBITDA is ₹70 cr, that's ~2.1x — comfortable. Lend up to ~₹150 cr.

Example 2 — tenor mismatch. A toll road with 20-year concession cash flows is offered a 5-year bullet. Cash flow easily covers interest, but there's no way to repay the principal in year 5 except refinancing — huge refinancing risk. *Restructure* to a long-tenor, sculpted amortization matching the concession.

Example 3 — sculpting to hold DSCR. A project's cash flow ramps from ₹30 cr to ₹90 cr over its life. Straight-line principal would breach DSCR in the early low-cash years. *Sculpt* principal so debt service tracks cash flow, keeping DSCR ~1.3x throughout.

How it is tested in interviews

- **"How would you size a loan?"** — "From stressed cash flow: max debt service = CADS ÷ required DSCR, then back out the principal that service supports at the given rate and tenor, cross-checked against a leverage cap."
- **"How do you decide the repayment structure?"** — "Match it to the cash-flow profile and asset life: front-load amortization for lumpy/declining cash flow, sculpt for ramping projects, avoid tenor mismatches and unnecessary bullets."
- **"Why is a bullet riskier than an amortizing loan?"** — "It concentrates repayment at maturity, so it relies on refinancing or a lump of cash being available then — refinancing risk."

- **"What's the danger of a tenor mismatch?"** — "Funding a long-life asset with short debt forces repeated refinancing; funding a short-life asset with long debt leaves debt outstanding after the asset stops earning."

Traps & common mistakes

- Sizing off **base-case** rather than **stressed** cash flow.
- **Tenor mismatch** with asset/cash-flow life.
- Over-reliance on **bullets** (refinancing risk).
- Ignoring the **leverage cap** cross-check — DSCR can look fine at dangerous leverage if rates are temporarily low.
- Structuring without covenants — no early warning or control.

First-principles recap

- Debt capacity is set by **stressed** cash flow and a required DSCR cushion.
- Max debt = f(max serviceable debt service, rate, tenor), capped by sector leverage.
- **Match amortization and tenor to the cash-flow profile and asset life.**
- Bullets create refinancing risk; sculpting holds DSCR steady.
- Structure and covenants convert a credit view into a safe transaction.

Quick-reference

Item	Rule
Max debt service	CADS ÷ required DSCR
Max debt	PV of that service at rate & tenor
Tenor	≤ economic life of asset/cash flow
Amortization	Match to cash-flow shape
Cross-check	Debt/EBITDA within sector cap

Covenants & Loan Documentation

The Problem / Why this matters

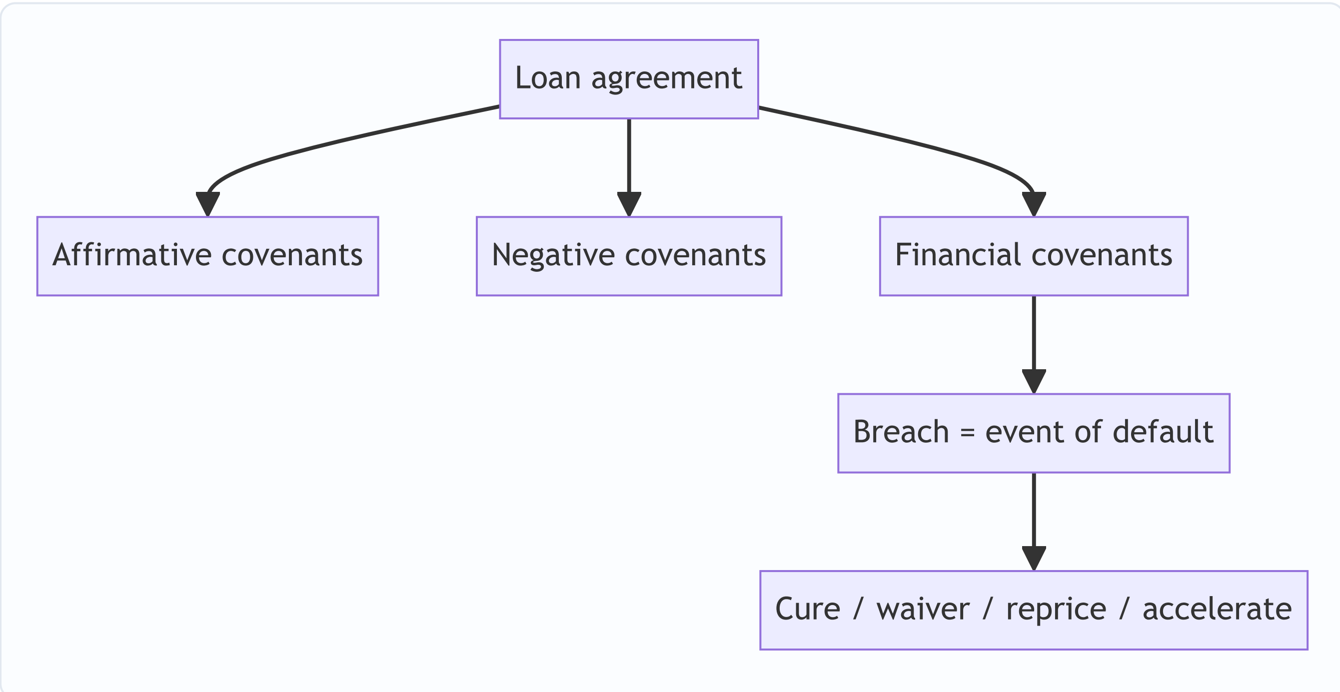
Once money is lent, the borrower controls the business and the lender is a passenger — until something goes wrong, by which point it may be too late to act. **Covenants are the contractual controls that give the lender early warning and the right to intervene** before a struggling borrower burns through the cash and assets that back the loan. They are the difference between finding out about trouble in time to protect your position and finding out when the borrower stops paying.

Core Idea

Covenants are promises in the loan agreement that constrain the borrower's behaviour and require it to maintain agreed financial thresholds. Breach them and the lender gains rights — to renegotiate, reprice, demand cure, or accelerate the loan. They convert a passive loan into an actively monitored, controllable exposure.

Why it works this way

The lender's protection is only as good as its information and its ability to act early. Financial covenants act as tripwires that fire while the borrower still has value to protect; restrictive covenants stop the borrower from taking actions (extra debt, asset sales, big dividends) that would strip that value away from creditors.



Full technical content

Three families of covenants: | Type | Purpose | Examples | |---|---|---| | **Affirmative** | Things the borrower must do | Provide audited statements on time, maintain insurance, pay taxes, keep assets in good order | | **Negative (restrictive)** | Things the borrower can't do without consent | Incur additional debt, grant liens, sell assets, pay dividends above a limit, make acquisitions, change control | | **Financial** | Maintain agreed metrics | Max leverage, min DSCR/ICR, min net worth, max capex |

Maintenance vs incurrence covenants: - **Maintenance** — tested every period; the borrower must *stay* within limits (e.g., Debt/EBITDA ≤ 4.0x each quarter). Common in bank loans. Fire early. - **Incurrence** — tested only when the borrower takes an *action* (issuing new debt, paying a dividend). Common in high-yield bonds; looser, borrower-friendly.

Typical financial covenant package (bank term loan): - Leverage: Debt/EBITDA ≤ a stepping-down cap. - Coverage: DSCR ≥ 1.2–1.5x; interest cover ≥ a floor. - Minimum tangible net worth. - Maximum annual capex. - Sometimes a minimum liquidity/cash balance.

Headroom & step-downs. Covenants are set with **headroom** above the base case (e.g., covenant leverage 4.5x vs projected 3.5x) so ordinary volatility doesn't trip them, and often **step down** over time as the borrower is expected to deleverage.

Events of default (EoD). Breach of a covenant, non-payment, cross-default (default on other debt triggers this one), insolvency, material adverse change. On EoD the lender can waive, agree a cure period, reprice for the extra risk, tighten terms, or **accelerate** (demand immediate repayment) and enforce security.

Security & guarantees (documented alongside): fixed and floating charges over assets, pledge of shares, personal/corporate guarantees, and the inter-creditor agreement governing ranking among lenders.

Worked examples

Example 1 — a maintenance covenant firing early. Loan has Debt/EBITDA ≤ 4.0x tested quarterly. EBITDA dips and leverage rises to 4.3x. The borrower is still paying, but the breach triggers a conversation *now*: the lender can require an equity cure, tighten terms, or reprice — months before any missed payment. That early warning is the whole point.

Example 2 — negative covenant blocking value leakage. A struggling borrower wants to pay a large dividend to the promoter. A restricted-payments covenant blocks it without lender consent, keeping cash inside the business to service debt. Without it, cash walks out the door ahead of creditors.

Example 3 — cross-default. A borrower defaults on a bond held by another lender. The cross-default clause in your loan lets you treat it as your default too, so you're not left waiting while other creditors act. It keeps lenders on an equal footing.

How it is tested in interviews

- **"What are covenants and why do they matter?"** — "Contractual controls that give the lender early warning and the right to act before a struggling borrower destroys value. Financial covenants are tripwires; restrictive covenants stop value leakage."
- **"Maintenance vs incurrence covenants?"** — "Maintenance are tested every period and fire early (bank loans); incurrence are tested only on an action like issuing debt (high-yield bonds), so they're looser."
- **"Give examples of financial covenants."** — Max Debt/EBITDA, min DSCR/ICR, min net worth, max capex.
- **"What happens on a covenant breach?"** — "It's an event of default: the lender can waive, require a cure, reprice, tighten terms, or accelerate and enforce security."

Traps & common mistakes

- Confusing **maintenance** (periodic, early-firing) with **incurrence** (action-triggered).
- Setting covenants with **no headroom** (constant technical breaches) or **too much** (never fire).
- Forgetting **cross-default** — it's what keeps you level with other creditors.
- Treating a breach as automatic acceleration — usually it opens options (cure/waiver/reprice), not instant repayment.
- Ignoring restrictive covenants' role in **preventing value leakage** (dividends, extra debt, asset sales).

First-principles recap

- Covenants give the lender **information and control** — early warning plus the right to act.
- Affirmative (must do), negative (can't do), financial (must maintain).
- Maintenance = periodic tripwires; incurrence = action-triggered.
- Breach = event of default → waive / cure / reprice / accelerate.
- Set with headroom and step-downs; back with security and cross-default.

Quick-reference

Concept	One-liner
Affirmative	Borrower must do (report, insure)
Negative	Borrower can't (extra debt, dividends, asset sales)
Financial	Maintain leverage/coverage/net worth
Maintenance	Tested each period, fires early
Incurrence	Tested on an action, looser
EoD remedies	Waive, cure, reprice, accelerate, enforce

Credit Ratings & the Agencies

The Problem / Why this matters

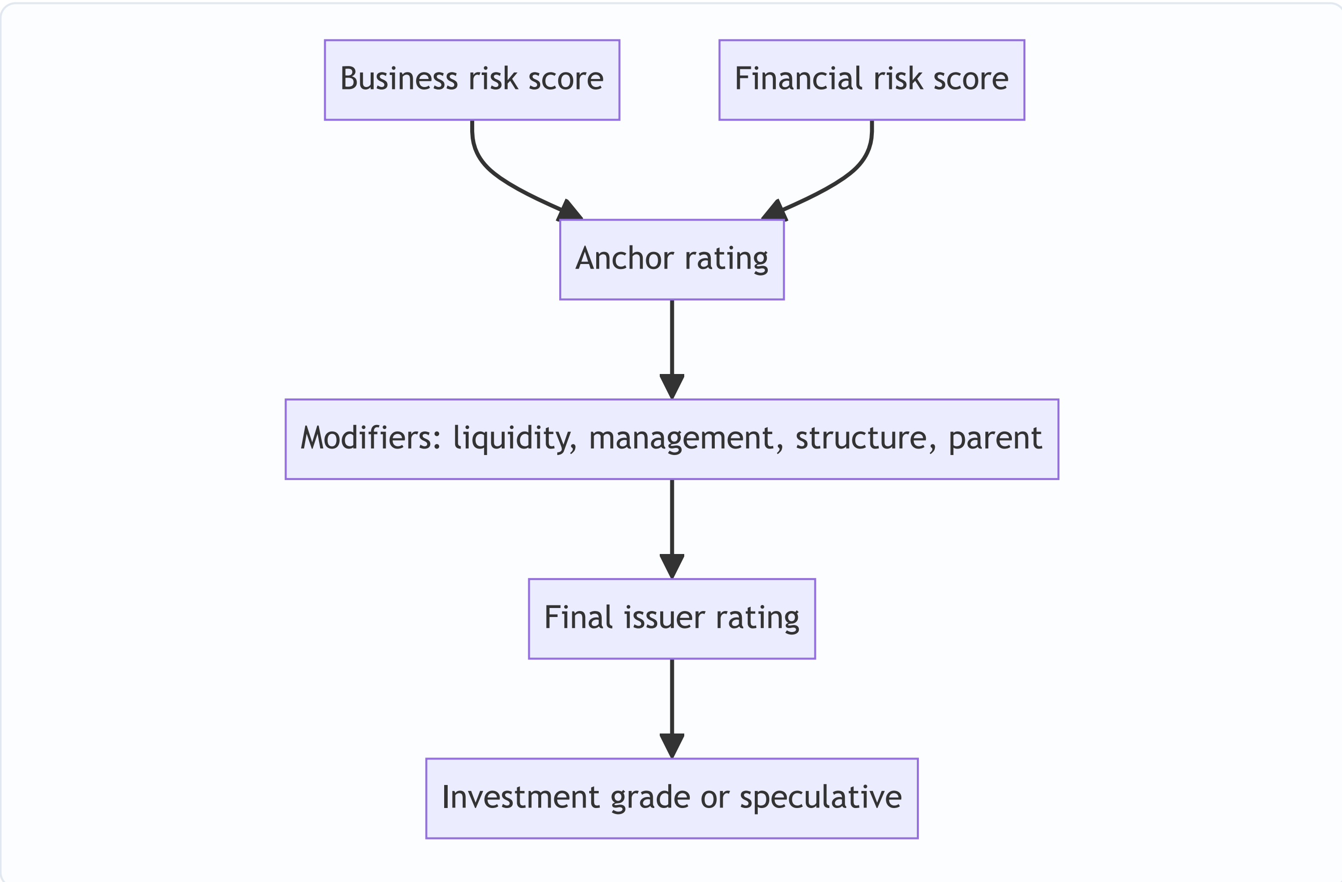
Investors buying thousands of bonds cannot spread every issuer themselves. They need an independent, comparable, standardized opinion of creditworthiness — and that is what **credit rating agencies** provide. Ratings drive who can buy a bond, how much capital a bank must hold against it, and the spread the issuer pays. A single-notch downgrade can raise borrowing costs by tens of basis points and, at the investment-grade boundary, force a wave of forced selling. Every credit interview expects you to know the scale and what moves a rating.

Core Idea

A **credit rating** is an agency's opinion of the relative likelihood that an issuer will default on its obligations (and, for some scales, the expected loss). It compresses a full business- and financial-risk analysis into a single symbol on a comparable scale, letting investors price and compare credit across the whole market.

Why it works this way

Ratings solve an information and comparability problem at scale. By using a consistent methodology across issuers, an agency lets an investor treat a BBB Indian manufacturer and a BBB European utility as broadly comparable in default risk, even though their businesses are nothing alike. That comparability is what makes bond markets liquid.



Full technical content

The rating scale (S&P / Fitch style; Moody's in brackets):

Grade	Ratings	Meaning
Investment grade	AAA (Aaa), AA, A, BBB (Baa)	Low-to-moderate default risk
Speculative / "junk"	BB (Ba), B, CCC (Caa), CC, C	Higher default risk
Default	D	In default

The critical line is **BBB−/Baa3 (the lowest investment grade) vs BB+/Ba1 (the highest speculative)**. Crossing below it ("fallen angel") can force index and mandate-driven investors to sell, widening spreads sharply. Notches within a grade are shown with +/− (S&P/Fitch) or 1/2/3 (Moody's).

Indian agencies: CRISIL, ICRA, CARE, India Ratings (Fitch), Acuité, Brickwork — they use an equivalent scale prefixed for the market (e.g., CRISIL AAA, AA, A, BBB...). SEBI regulates them.

How a rating is built (agency methodology): 1. **Business risk** — industry risk + competitive position (country risk overlaid). 2. **Financial risk** — leverage, coverage, cash flow, financial policy. 3. Combine into an **anchor** rating via a matrix. 4. **Modifiers** — diversification, liquidity, management/governance, capital structure, and parent/group support (notching up or down). 5. Result: the **issuer credit rating**; individual instruments may be notched for seniority/security.

Issuer vs issue rating. The issuer rating reflects overall default risk; a specific bond may be rated higher (senior secured) or lower (subordinated) based on expected recovery.

Through-the-cycle vs point-in-time. Agency ratings aim to be **through-the-cycle** (stable, not reacting to every quarter), whereas market spreads and internal PD models are more point-in-time.

Rating triggers & watch. Bonds/loans may have rating-linked terms (step-up coupons, covenants). Agencies signal likely changes via "outlook" (positive/negative/stable) and "credit watch."

Worked examples

Example 1 — the investment-grade cliff. A firm rated BBB− (lowest IG) is downgraded to BB+ (highest speculative). Nothing about the business changed dramatically — but many funds and indices can only hold IG, so they must sell. Forced selling widens the spread far more than the one-notch move suggests, and the firm's future borrowing cost jumps. *This is why the IG boundary matters more than any other notch.*

Example 2 — issue notching. An issuer is rated BB. Its senior secured loan, with strong collateral and high expected recovery, is notched up to BB+; its subordinated bond, with low recovery, is notched down to BB−. Same issuer, different instrument ratings driven by seniority and security.

Example 3 — parent support. A standalone subsidiary would be rated BB, but it's strategically core to a AAA parent that is highly likely to support it. The agency notches the subsidiary up to, say, A− reflecting expected parental support. Remove the support assumption and the rating falls.

How it is tested in interviews

- **"What's the investment-grade cutoff?"** — "BBB− / Baa3. Below that is speculative or high-yield. Crossing it can force index/mandate selling and widen spreads sharply."
- **"How is a rating determined?"** — "Business-risk and financial-risk scores combine into an anchor, then modifiers (liquidity, management, group support) adjust it to the final rating."
- **"What does a downgrade do?"** — "Raises borrowing cost, widens spreads, can trigger rating-linked covenants or step-ups, and at the IG boundary forces selling."

- **"Issuer vs issue rating?"** — "Issuer reflects overall default risk; a specific instrument is notched up or down for its seniority and expected recovery."

Traps & common mistakes

- Not knowing the **BBB−/Baa3** boundary and why it matters.
- Treating a rating as a **default probability** — it's a relative, ordinal opinion, not a precise PD.
- Ignoring **outlook/watch** signals of a pending change.
- Confusing **issuer** and **issue** ratings.
- Forgetting **group/parent support** can drive a subsidiary's rating.

First-principles recap

- A rating is a standardized, comparable opinion of relative default risk.
- Investment grade (AAA–BBB−) vs speculative (BB+ and below); the IG boundary is decisive.
- Built from business + financial risk → anchor → modifiers → final rating.
- Issue ratings notch off the issuer for seniority/recovery.
- Downgrades raise cost, widen spreads, and can force selling.

Quick-reference

Item	Note
IG range	AAA to BBB− (Baa3)
Speculative	BB+ (Ba1) and below
Key boundary	BBB−/BB+ (fallen angel)
Indian agencies	CRISIL, ICRA, CARE, India Ratings
Build	Business + financial risk → anchor → modifiers
Issue vs issuer	Notch for seniority/recovery

Default Probability: Altman Z & Merton

The Problem / Why this matters

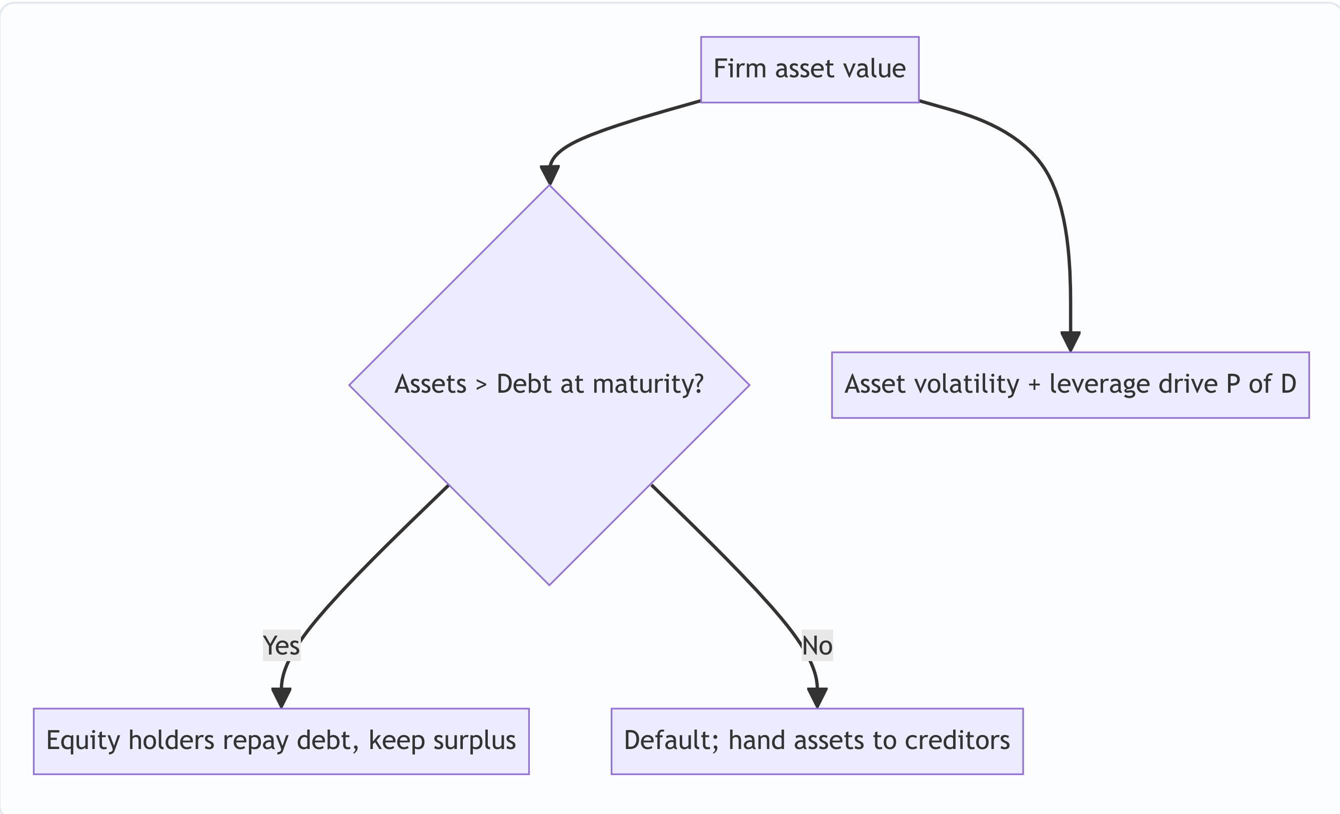
Ratios and judgement give a qualitative sense of risk, but modern credit needs a **number** — the probability that this borrower defaults. Quantifying default lets you price loans, compute expected loss, hold the right capital, and rank a whole portfolio consistently. Two classic models sit behind almost every default-probability conversation: **Altman's Z-score** (an accounting-ratio model) and the **Merton structural model** (an options-based model). Interviewers love both because they test whether you understand what actually drives default.

Core Idea

- **Altman Z-score** — a weighted combination of five accounting ratios that classifies a firm into safe, grey, or distress zones. Simple, transparent, ratio-based.
- **Merton model** — treats a firm's equity as a **call option on its assets**; default happens when asset value falls below the debt due. It links default to asset volatility and leverage, and underpins market-based PD (e.g., Moody's KMV).

Why it works this way

Default is fundamentally about **assets falling below liabilities** (or cash falling below obligations). Altman captures this empirically through ratios that historically separated defaulters from survivors. Merton captures it structurally: equity holders own the upside and can "walk away" if assets fall below debt, exactly like a call option — so option maths gives the default probability.



Full technical content

Altman Z-score (original, manufacturing): Z = 1.2·X1 + 1.4·X2 + 3.3·X3 + 0.6·X4 + 1.0·X5, where - X1 = Working Capital / Total Assets (liquidity) - X2 = Retained Earnings / Total Assets (cumulative profitability/age) - X3 = EBIT / Total Assets (operating profitability) - X4 = Market Value of Equity / Total Liabilities (leverage/cushion) - X5 = Sales / Total Assets (asset turnover)

Zones: **Z > 2.99 = safe; 1.81–2.99 = grey; Z < 1.81 = distress.** Variants exist for private firms (Z′, using book equity) and non-manufacturers (Z″, dropping X5). It's a scorecard: transparent, quick, but backward-looking and accounting-based.

Merton structural model (intuition + mechanics): - Firm assets V follow a random walk with volatility σ_V; debt face value D is due at time T. - Equity = a call option on assets with strike D: max(V − D, 0) at T. - **Distance to default (DD)** ≈ (V − D) / (V·σ_V) — how many standard deviations of asset value sit between today's assets and the default point. - **PD** = probability assets end below D = N(−DD) (a normal-distribution tail). - Higher leverage (lower V/D) and higher asset volatility (σ_V) → smaller DD → higher PD.

The insight: **default risk rises with leverage AND with asset volatility.** Two firms at the same leverage differ in default risk if their asset volatility differs — which ties straight back to business risk.

Structural vs reduced-form vs scorecards: | Approach | Basis | Used for | |---|---|---| | Structural (Merton/KMV) | Asset value & volatility from equity market | Listed firms, market-based PD | | Reduced-form | Default as a random event calibrated to spreads | Pricing credit derivatives | | Scorecards (Altman, internal) | Weighted ratios / logistic regression | Private firms, banks' internal ratings |

Worked examples

Example 1 — Altman Z. A manufacturer: WC/TA 0.15, RE/TA 0.20, EBIT/TA 0.10, MVE/TL 0.80, Sales/TA 1.10. Z = 1.2(0.15) + 1.4(0.20) + 3.3(0.10) + 0.6(0.80) + 1.0(1.10) = 0.18 + 0.28 + 0.33 + 0.48 + 1.10 = **2.37** → grey zone: watch closely, not yet distress.

Example 2 — distance to default. Assets ₹1,000 cr, debt due ₹700 cr, asset volatility 25%. DD ≈ (1000 − 700)/(1000 × 0.25) = 300/250 = **1.2 standard deviations**. PD ≈ N(−1.2) ≈ **11.5%**. If asset volatility rose to 40%, DD ≈ 300/400 = 0.75 → PD ≈ N(−0.75) ≈ **22.7%** — same leverage, double the default risk from higher volatility.

Example 3 — leverage effect. Same assets ₹1,000 cr and σ 25%, but debt rises to ₹850 cr. DD ≈ 150/250 = 0.6 → PD ≈ N(−0.6) ≈ **27%**. More debt shrinks the cushion and raises PD sharply.

How it is tested in interviews

- **"How would you estimate probability of default?"** — "Scorecards like Altman's Z for private firms, or a structural Merton/KMV model for listed firms where equity market data gives asset value and volatility, plus internal logistic-regression models."
- **"Explain the Merton model."** — "Equity is a call option on the firm's assets; default is when assets fall below debt at maturity. PD depends on distance to default, which falls with higher leverage and higher asset volatility."
- **"What drives default risk in Merton?"** — "Leverage and asset volatility. Two equally-levered firms differ in PD if their asset volatility differs."

- **"What is the Altman Z-score?"** — Five weighted ratios; $Z < 1.81$ distress, > 2.99 safe; quick, transparent, but accounting-based and backward-looking.

Traps & common mistakes

- Treating Altman Z as precise — it's a **classifier**, and coefficients are dated/sector-specific.
- Forgetting Merton needs **asset** value/volatility, inferred from **equity** (only works well for listed firms).
- Ignoring that **volatility**, not just leverage, drives PD.
- Confusing **structural** (asset-based) and **reduced-form** (spread-calibrated) models.

First-principles recap

- Default ≈ assets falling below debt; models quantify that probability.
- **Altman Z** = weighted accounting ratios → safe/grey/distress zones.
- **Merton** = equity as a call on assets; PD = N(−distance to default).
- PD rises with **leverage** and **asset volatility** — volatility ties to business risk.
- Structural for listed firms, scorecards for private, reduced-form for pricing derivatives.

Quick-reference

Model	Formula / rule
Altman Z	$1.2X1+1.4X2+3.3X3+0.6X4+1.0X5$; <1.81 distress
Distance to default	$(V - D)/(V \cdot \sigma\_V)$
Merton PD	$N(-DD)$
Drivers	Leverage ↑, asset volatility ↑ → PD ↑

PD, LGD, EAD & Expected Loss

The Problem / Why this matters

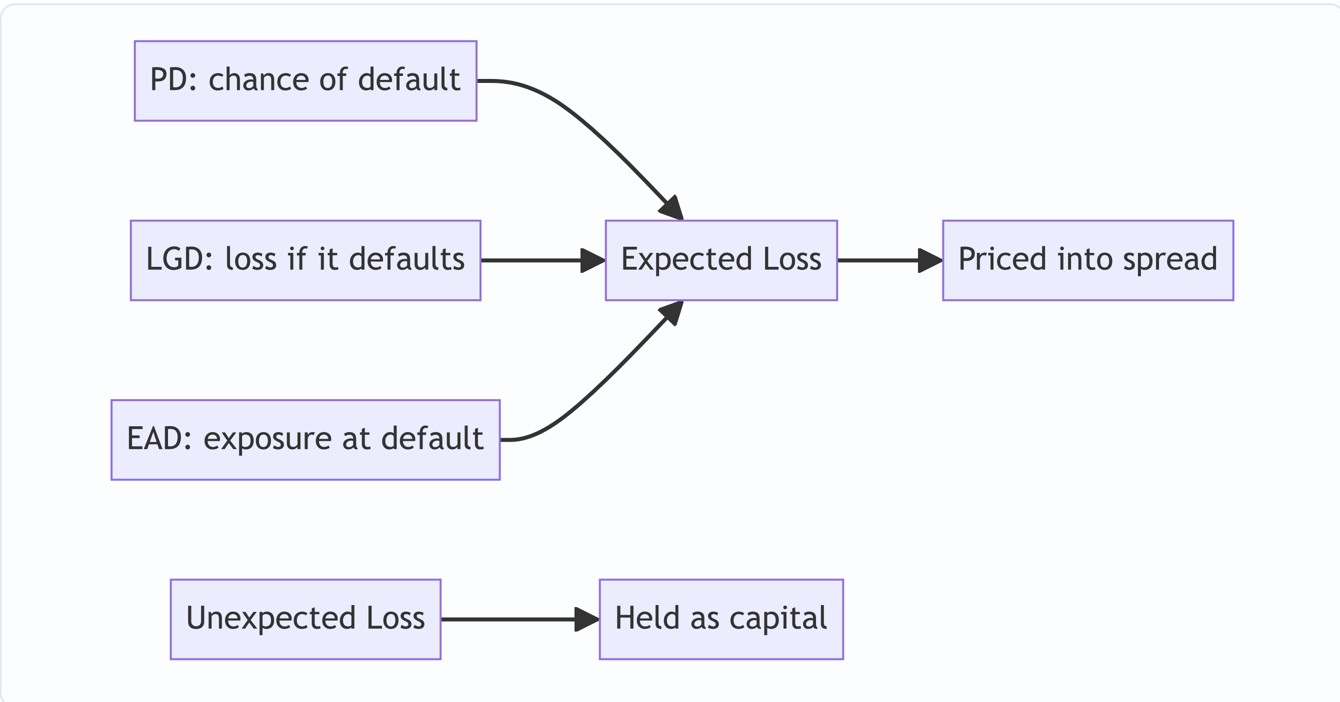
Knowing a borrower *might* default isn't enough to price a loan or hold capital against it. You need to turn "risky" into a rupee number: **how much do I expect to lose on this exposure, on average, over a year?** That number — expected loss — is the foundation of loan pricing, provisioning (IFRS 9), and regulatory capital (Basel). It decomposes into three intuitive pieces: how likely default is, how much you lose if it happens, and how much is exposed at that moment.

Core Idea

Expected Loss (EL) = PD × LGD × EAD. - **PD** (Probability of Default) — chance the borrower defaults over the horizon (usually 1 year). - **LGD** (Loss Given Default) — the fraction of exposure you *lose* if it defaults = 1 − recovery rate. - **EAD** (Exposure at Default) — how much will actually be outstanding when default occurs.

Why it works this way

Loss only happens if the borrower defaults (PD), and even then you recover some of it (so you lose only LGD), applied to whatever is actually drawn at that time (EAD). Multiplying the three gives the average, or *expected*, loss. Because it's an average, you price it into the spread as a cost of doing business; the *unexpected* loss (volatility around the average) is what capital is held against.



Full technical content

The three parameters: | Parameter | Definition | Drivers | |---|---|---| | **PD** | Probability of default over the horizon | Rating, financials, models (Altman/Merton/scorecard) | | **LGD** | 1 − recovery rate | Seniority, security/collateral, structure, industry | | **EAD** | Expected outstanding at default | Current drawn + likely further drawdowns on committed lines |

Expected vs unexpected loss: - **Expected loss (EL)** = PD × LGD × EAD — the *average* loss; covered by **provisions** and **pricing** (built into the spread). - **Unexpected loss (UL)** — the volatility of losses around EL (many borrowers defaulting together in a downturn); covered by **economic/regulatory capital**. You price EL; you hold capital for UL.

Point-in-time vs through-the-cycle PD. PIT PD reflects current conditions (rises in recessions); TTC PD is a cycle-average (more stable). Basel and IFRS 9 use different flavours; rating-based PDs are more TTC.

EAD nuance. For a term loan fully drawn, EAD ≈ outstanding balance. For a **revolving/committed line**, borrowers tend to **draw more as they approach default** (drawing on remaining availability while they still can), so EAD > current drawn; modelled with a **credit conversion factor (CCF)** on the undrawn amount.

Link to provisioning and capital: - **IFRS 9 ECL** = PD × LGD × EAD (discounted), staged 1/2/3 as credit deteriorates (12-month ECL in stage 1, lifetime ECL in stages 2–3). - **Basel** uses PD, LGD, EAD in the internal-ratings-based (IRB) capital formula, where capital is sized to *unexpected* loss at a high confidence level.

Worked examples

Example 1 — expected loss. Exposure ₹100 cr, PD 2%, recovery 40% (so LGD 60%). EL = 0.02 × 0.60 × 100 = **₹1.2 cr**. That's the average annual loss — it should be covered by charging at least ~120 bps of spread on the ₹100 cr just for credit cost, before profit and capital.

Example 2 — LGD from structure. Same borrower (PD 2%, EAD ₹100 cr), two facilities: a senior secured loan with 70% recovery (LGD 30%) → EL = 0.02 × 0.30 × 100 = **₹0.6 cr**; a subordinated bond with 20% recovery (LGD 80%) → EL = 0.02 × 0.80 × 100 = **₹1.6 cr**. Same default risk, very different loss because of seniority/security — which is why the sub bond pays a wider spread.

Example 3 — EAD on a revolver. A ₹100 cr committed line is currently ₹40 cr drawn. As default nears, the borrower draws more; with a 60% CCF on the ₹60 cr undrawn, EAD = 40 + 0.6 × 60 = **₹76 cr**. Using current drawn (₹40 cr) would badly understate exposure and expected loss.

How it is tested in interviews

- **"How do you calculate expected loss?"** — "PD × LGD × EAD. Probability of default, times loss given default (1 − recovery), times exposure at default."
- **"What's the difference between expected and unexpected loss?"** — "Expected loss is the average, priced into the spread and covered by provisions. Unexpected loss is the volatility around it and is covered by capital."
- **"Two bonds, same issuer, different spreads — why?"** — "Same PD but different LGD: the senior secured one recovers more (lower LGD, tighter spread); the subordinated one recovers less."
- **"Why can EAD exceed current drawings?"** — "On committed lines, borrowers draw down further as they approach default; we model that with a credit conversion factor."

Traps & common mistakes

- Forgetting **LGD = 1 − recovery**, not the recovery itself.
- Using **current drawn** as EAD on a revolver (ignoring drawdown behaviour).

- Confusing **expected** loss (priced/provisioned) with **unexpected** loss (capital).
- Assuming PD, LGD, EAD are independent — in downturns they all worsen together (correlation).
- Mixing **PIT and TTC** PDs without stating which.

First-principles recap

- **EL = PD × LGD × EAD** — the average loss, priced into the spread.
- **LGD = 1 − recovery**, driven by seniority and security.
- **EAD** can exceed current drawings on committed lines (CCF).
- Expected loss → pricing/provisions; unexpected loss → capital.
- IFRS 9 ECL and Basel IRB both build on these three parameters.

Quick-reference

Term	Definition
PD	Probability of default (horizon)
LGD	1 − recovery rate
EAD	Exposure at default (drawn + CCF × undrawn)
EL	PD × LGD × EAD
Covered by	EL → spread/provisions; UL → capital

Recovery, Seniority & the Capital-Structure Waterfall

The Problem / Why this matters

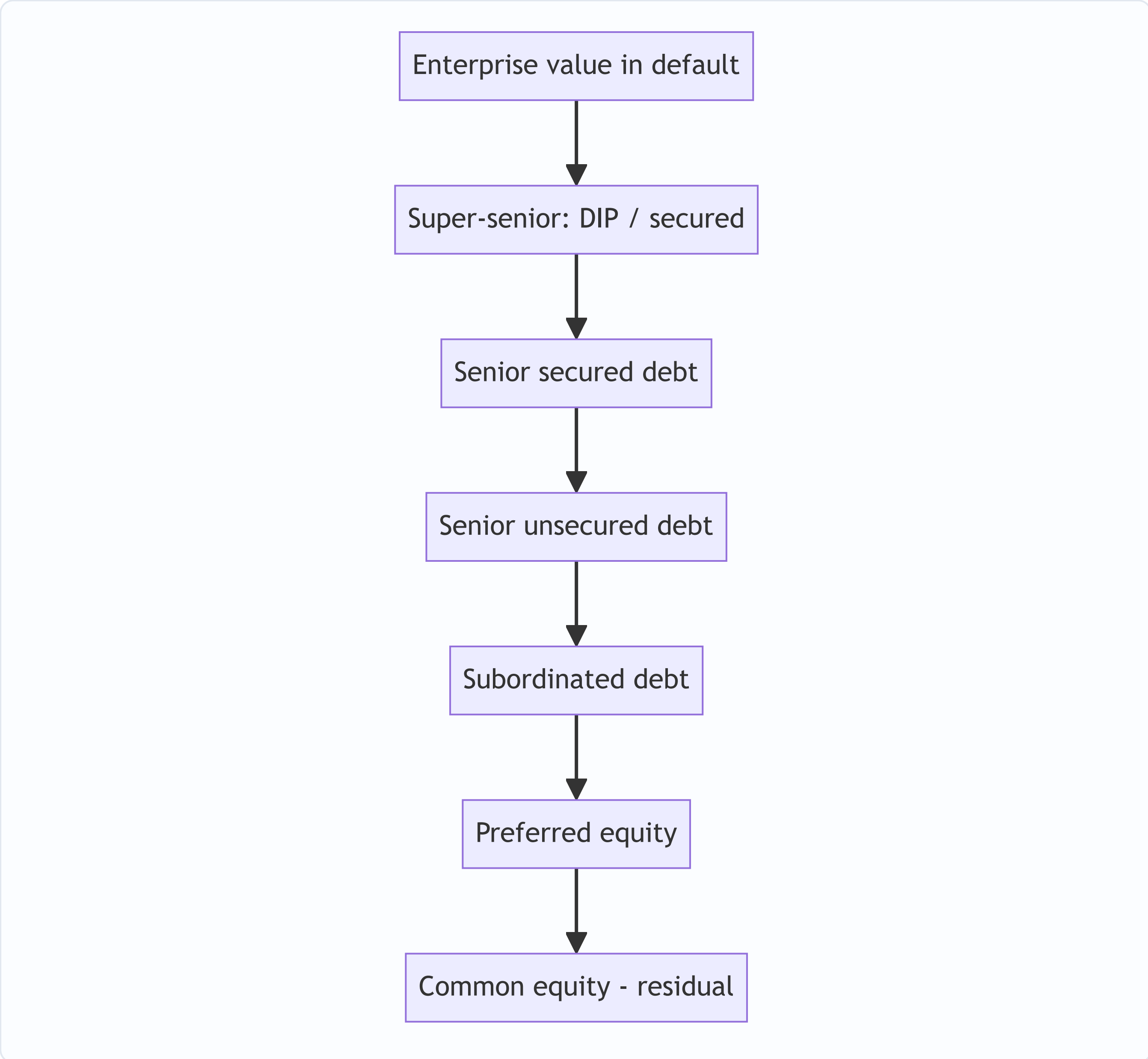
When a borrower defaults, there usually *is* value left — just not enough to pay everyone. Who gets paid, in what order, and how much is left over for you depends entirely on **where your claim sits in the capital structure** and whether it is secured. Two lenders to the same defaulted company can recover 90% and 10% respectively purely because of seniority and security. Understanding the **waterfall** is what lets you estimate LGD, price subordinated debt, and structure a facility to protect your recovery.

Core Idea

In default, a company's value is distributed to claimants in a strict **priority order** — the waterfall. Secured and senior claims are paid first (high recovery, low LGD); subordinated and equity claims are paid last, often getting little or nothing. **Recovery is driven by seniority, security, and the value available**, and LGD = 1 − recovery.

Why it works this way

Debt is a contract with an agreed rank. Lenders accept lower returns for higher priority; subordinated lenders and equity accept lower priority for higher expected return. In bankruptcy, the **absolute priority rule** enforces that senior claims are paid in full before junior claims receive anything — that's the deal each layer signed up for.



Full technical content

The priority waterfall (typical order of payment): | Rank | Claim | Typical recovery | |---|---|---| | 1 | Administrative/insolvency costs, super-priority (DIP) financing | High | | 2 | **Senior secured** (fixed/floating charge over specific assets) | 60–90% | | 3 | **Senior unsecured** | 30–50% | | 4 | **Subordinated / junior** debt | 10–30% | | 5 | Preferred equity | Low | | 6 | **Common equity** (residual) | Often ~0 |

(Statutory dues — employee wages, taxes — often rank ahead in many jurisdictions; India's IBC has its own waterfall under Section 53.)

What drives recovery (and thus LGD): - **Seniority** — higher rank, more of the value. - **Security/collateral** — a specific charge on quality assets means you're paid from those assets first; quality and liquidity of the collateral matter. - **Enterprise value in default** — a viable business sold as a going concern recovers far more than a liquidation of assets at fire-sale prices. - **Industry & asset type** — tangible-asset-heavy firms (real estate, utilities) recover more than asset-light ones (services, tech). - **Structural position** — debt at an operating subsidiary (closer to the assets) outranks debt at a holding company (**structural subordination**).

Secured vs unsecured, fixed vs floating charge: a **fixed charge** attaches to specific assets (best protection); a **floating charge** covers a changing pool (inventory, receivables) and crystallizes on default. Secured lenders are paid from their collateral first; any shortfall ranks as unsecured for the balance.

Going concern vs liquidation. Restructuring/resolution that keeps the business running (a going-concern sale) usually preserves more value than piecemeal liquidation — which is the logic behind modern insolvency regimes (Chapter 11 in the US, the IBC in India) that favour resolution over liquidation.

Worked examples

Example 1 — the waterfall in action. A defaulted firm's assets realize ₹600 cr. Claims: senior secured ₹400 cr, senior unsecured ₹300 cr, subordinated ₹200 cr. - Senior secured: paid in full ₹400 cr → **recovery 100%, LGD 0%.** - Remaining ₹200 cr to senior unsecured (₹300 cr claim): **recovery 67%, LGD 33%.** - Subordinated: nothing left → **recovery 0%, LGD 100%.** Same company, recoveries of 100% / 67% / 0% purely by rank.

Example 2 — collateral quality. Two senior secured lenders. Lender A holds a fixed charge on prime real estate (liquid, holds value) → ~85% recovery. Lender B holds a floating charge on specialised, fast-depreciating machinery → ~40% recovery. Same seniority label, very different LGD because collateral quality differs.

Example 3 — structural subordination. HoldCo bondholders and OpCo lenders both lend ₹100 cr. The assets sit in the OpCo. On default, OpCo lenders are paid from the OpCo assets first; HoldCo bonds only receive whatever is left *after* OpCo creditors are satisfied — effectively subordinated despite being "senior" on paper. *Lesson:* lend as close to the assets as possible.

How it is tested in interviews

- **"What determines recovery in a default?"** — "Seniority, security/collateral quality, the enterprise value available (going concern vs liquidation), and structural position — how close your claim is to the assets."
- **"Explain the capital-structure waterfall."** — Admin/super-senior → senior secured → senior unsecured → subordinated → preferred → common equity, paid in strict priority (absolute priority rule).
- **"Two bonds, same issuer, 80% vs 20% recovery — why?"** — "Different seniority/security: the senior secured is paid from collateral first; the subordinated gets the residual."
- **"What is structural subordination?"** — "HoldCo debt ranks behind OpCo creditors because the assets and operating creditors sit at the OpCo, which is paid first."

Traps & common mistakes

- Assuming "senior" always recovers well — **collateral quality** and **enterprise value** matter as much as rank.
- Ignoring **structural subordination** (HoldCo vs OpCo).
- Forgetting **statutory dues** (wages, taxes) can rank ahead.
- Confusing **going-concern** value with liquidation value — they can differ hugely.
- Treating recovery as fixed — it varies with the cycle (recoveries fall in downturns, when defaults rise).

First-principles recap

- Default value is distributed by a strict **priority waterfall** (absolute priority rule).
- Recovery is driven by **seniority, security, enterprise value, and structural position.**
- LGD = 1 − recovery; senior secured → low LGD, subordinated → high LGD.
- Collateral quality and going-concern vs liquidation swing recoveries widely.
- Lend as **close to the assets** as possible (avoid structural subordination).

Quick-reference

Rank	Claim	Recovery
1	Admin / super-senior (DIP)	Highest
2	Senior secured	60–90%
3	Senior unsecured	30–50%
4	Subordinated	10–30%
5–6	Preferred / common equity	~0
LGD	= 1 − recovery	

Working-Capital & Trade-Finance Credit

The Problem / Why this matters

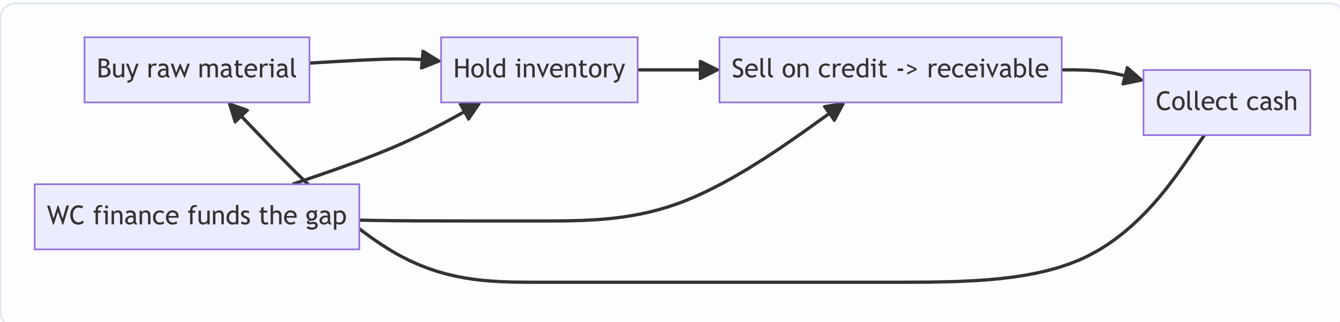
Not all lending funds long-term assets. A huge share of bank credit funds the **day-to-day operating cycle** — the gap between paying for inventory and collecting from customers — and the **risks of trade**, especially cross-border trade where buyer and seller don't trust each other. This is the bread-and-butter of commercial banking, and it uses a different toolkit from term lending: limits sized to the operating cycle, self-liquidating structures, and instruments like letters of credit and factoring. Interviewers for corporate/commercial banking and NBFC roles test it directly.

Core Idea

Working-capital finance funds the **current-asset cycle** (inventory + receivables, net of payables) and is repaid as that cycle turns over — it is *self-liquidating*. Trade finance de-risks specific transactions between counterparties who lack trust, using bank credit (letters of credit, guarantees) or receivables-based funding (factoring, forfaiting).

Why it works this way

A business must pay for raw materials and hold inventory and wait for customers to pay — that tied-up cash is working capital, and it grows with sales. Financing it against the underlying current assets is naturally self-liquidating: as inventory sells and receivables collect, the loan is repaid, then re-drawn for the next cycle. Trade finance works because a *bank's* promise to pay is more trusted than a distant buyer's.



Full technical content

Sizing working-capital limits. Limits track the operating cycle and the working-capital gap: - The **cash conversion cycle** = DIO + DSO − DPO (days). Longer cycle → more financing needed. - Bank methods: the **drawing-power** approach (advance a percentage against eligible current assets, e.g., 75% of receivables under 90 days + inventory less margin, minus creditors), or turnover/Nayak-committee style methods for SMEs. - Facility types: **cash credit / overdraft** (revolving, against a drawing-power base), **working-capital demand loan**, **bill discounting**.

Trade-finance instruments: | Instrument | What it does | |---|---| | **Letter of Credit (LC)** | Bank guarantees payment to the seller once compliant shipping documents are presented — replaces buyer's credit risk with the bank's | | **Bank Guarantee (BG)** | Bank promises to pay a beneficiary if the applicant fails to perform (performance/financial guarantee) | | **Factoring** | Seller sells receivables to a financier for immediate cash (with or without recourse) | | **Forfaiting** | Purchase of medium-term export receivables without recourse to the seller | | **Bill discounting** | Bank advances against a bill of exchange before its due date | | **Supply-chain finance** | Financing the payables/receivables between a large anchor and its suppliers |

LC mechanics (documentary credit): the buyer's bank (issuing bank) issues an LC in favour of the seller; the seller ships and presents documents; if documents comply, the bank pays regardless of the buyer's willingness. It shifts risk from the buyer to the (typically stronger) issuing bank, and is governed by UCP 600.

Recourse vs non-recourse factoring. With recourse, the seller bears the default risk if the customer doesn't pay; without recourse (and forfaiting), the financier takes the credit risk — priced accordingly.

Self-liquidating principle. The lender lends against assets that will turn into cash in the normal course (receivables collect, inventory sells), so the facility repays itself. Diversion of these funds to fixed assets ("diversion of working capital to capex") is a classic warning sign of stress.

Worked examples

Example 1 — sizing a cash-credit limit. A firm has average receivables ₹120 cr (advance 75% → ₹90 cr) and eligible inventory ₹80 cr (advance 60% → ₹48 cr), less creditors ₹30 cr. Drawing power = 90 + 48 − 30 = **₹108 cr**. The bank sanctions a cash-credit limit around this, adjusted each month as the current-asset base moves.

Example 2 — LC replacing buyer risk. An Indian exporter ships ₹5 cr of goods to a first-time overseas buyer it doesn't trust. The buyer opens an LC through a strong international bank. The exporter ships, presents compliant documents, and is paid by the bank — it never relied on the buyer's creditworthiness, only the bank's. Risk transformed from buyer to bank.

Example 3 — factoring for cash flow. A supplier with ₹50 cr of receivables due in 90 days needs cash now. It factors them without recourse at a 2% fee plus interest, receiving ~₹48.5 cr immediately. The financier now bears the customers' credit risk and collects at maturity. The supplier trades a small cost for immediate liquidity and offloaded risk.

How it is tested in interviews

- **"What is working-capital finance and how is it repaid?"** — "Funds the operating cycle — inventory and receivables net of payables — and is self-liquidating: it's repaid as inventory sells and receivables collect, then re-drawn."
- **"What is a letter of credit?"** — "A bank's promise to pay the seller on the buyer's behalf once compliant shipping documents are presented — it swaps the buyer's credit risk for the bank's."
- **"How would you size a cash-credit limit?"** — "Drawing power: a margin-adjusted advance against eligible receivables and inventory, less creditors; tracks the working-capital gap and cash conversion cycle."
- **"Recourse vs non-recourse factoring?"** — "With recourse the seller keeps the default risk; without recourse the financier takes it and charges more."
- **"What's a red flag in working-capital lending?"** — "Diversion of working-capital finance into fixed assets or promoter uses — the facility stops being self-liquidating."

Traps & common mistakes

- Treating working-capital limits as **term** debt — they should turn over, not fund fixed assets.
- Missing **diversion of funds** as a stress signal.
- Confusing an **LC** (payment undertaking on documents) with a **bank guarantee** (pays on non-performance).
- Ignoring **collateral/margin** on the drawing base (advancing 100% of receivables).
- Overlooking **recourse** terms in factoring — who actually bears the default risk.

First-principles recap

- Working-capital finance funds the operating cycle and is **self-liquidating**.
- Limits track the **cash conversion cycle** and drawing power against current assets.
- **LCs and BGs** substitute a bank's credit for a counterparty's in trade.
- **Factoring/forfaiting** turn receivables into immediate cash, shifting credit risk (with/without recourse).
- Diversion of working-capital funds to fixed assets is a classic warning sign.

Quick-reference

Item	Note
Cash conversion cycle	DIO + DSO − DPO
Drawing power	Margined advance vs receivables + inventory − creditors
LC	Bank pays seller on compliant docs (UCP 600)
BG	Bank pays on applicant's non-performance
Factoring	Sell receivables for cash (recourse/non-recourse)
Red flag	WC funds diverted to fixed assets

Bond & Fixed-Income Credit

The Problem / Why this matters

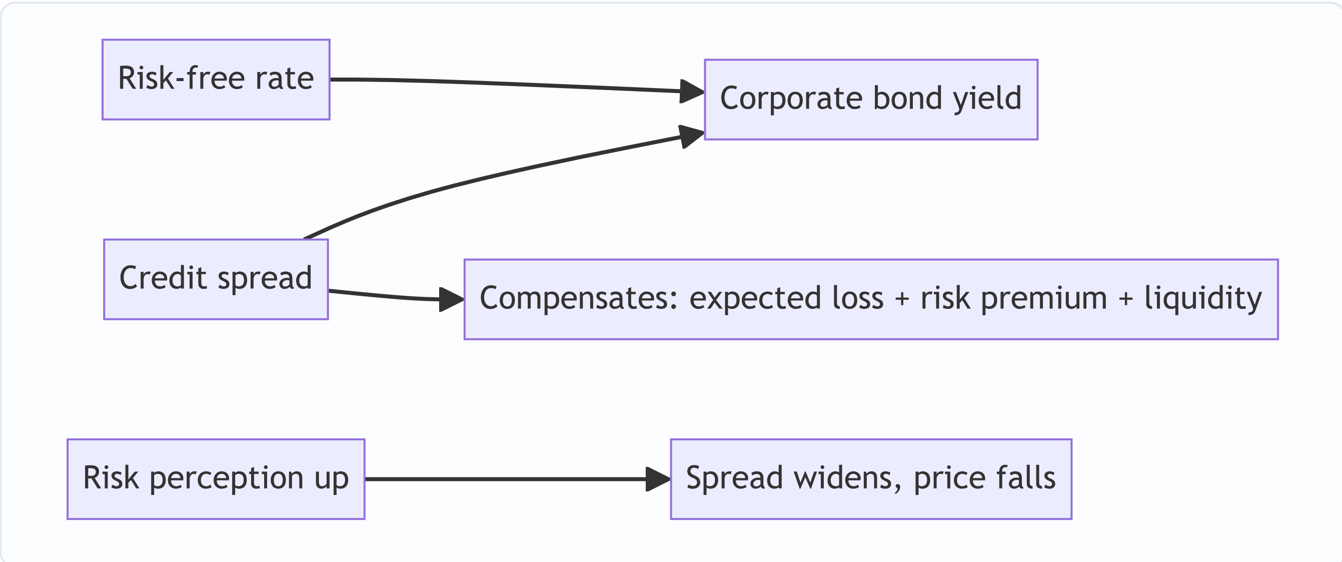
Credit isn't only bank loans — most large-company debt trades as **bonds** in public markets, where the price of credit risk is visible every second as the **credit spread**. For anyone in fixed income, credit research, or risk, you must understand what a spread is, what moves it, how ratings map to spreads, and how to find value in credit. This is where credit analysis meets market pricing.

Core Idea

A corporate bond's yield = a **risk-free rate** + a **credit spread**. The spread is the extra yield investors demand to bear the bond's default and liquidity risk. Credit investing is the business of judging whether that spread **over-compensates or under-compensates** for the actual risk — and how the spread will move.

Why it works this way

A lender to a risky borrower must be paid for the expected loss (PD × LGD) plus a premium for the *uncertainty* of that loss and for illiquidity. The market prices this as a spread over the risk-free curve. When perceived risk rises (weaker economy, worse fundamentals), required spread widens and bond prices fall; when risk falls, spreads tighten and prices rise.



Full technical content

Bond yield decomposition: Yield = risk-free rate + credit spread. The spread compensates for (i) **expected loss** (PD × LGD), (ii) a **risk premium** for uncertainty/undiversifiable credit risk, and (iii) **liquidity**.

Spread measures: | Measure | What it is | |---|---| | Nominal spread | Yield − matching-maturity govt yield | | **Z-spread** | Constant spread over the whole spot curve that prices the bond | | **OAS** (option-adjusted spread) | Z-spread adjusted for embedded options (calls/puts) | | Asset-swap spread | Spread over the swap curve via an asset swap | | CDS spread | Cost of default protection (a cleaner default read) |

Price–yield and spread duration. Bond price moves inversely to yield; a **1 bp spread widening** cuts price by roughly (spread duration × 1 bp). So credit P&L ≈ −spread duration × change in spread. Longer-dated, lower-coupon bonds have higher spread duration and more spread sensitivity.

Rating ↔ spread mapping. Lower ratings carry wider spreads: AAA a few tens of bps, BBB moderately more, BB/B several hundred bps, CCC very wide. The **investment-grade vs high-yield** divide (BBB−/BB+) is a step-change in spread and investor base. Spreads also embed a cycle: they widen in recessions (when defaults rise) and compress in booms.

Investment grade vs high yield: | | Investment grade | High yield | |---|---|---| | Rating | BBB− and above | BB+ and below | | Spread | Tight | Wide | | Driver | Rates + modest spread | Spread/default-dominated | | Covenants | Lighter (incurrence) | Tighter, more analysis |

Relative value / finding value in credit. Compare a bond's spread to (i) its fundamentals-implied fair spread, (ii) peers of the same rating/sector, and (iii) the same issuer's other bonds and its CDS (basis). Buy bonds whose spread over-pays for the risk; avoid those that under-pay. Watch the **credit cycle** — spreads mean-revert and gap wider in stress.

The credit-spread cycle. Spreads are tight and complacent late in an expansion, then gap violently wider in a downturn as defaults and risk aversion rise — the single biggest driver of credit returns.

Worked examples

Example 1 — yield build-up. 5-year G-Sec yields 7.0%. A BBB corporate of the same maturity trades at a 200 bp spread → yield **9.0%**. A BB peer trades at 450 bp → **11.5%**. The 250 bp gap between BBB and BB is the market's price for the extra default risk and the IG/HY divide.

Example 2 — spread P&L. You hold a bond with spread duration 6. Its spread widens 50 bp on weak results. Price impact ≈ −6 × 0.50% = **−3.0%**. Even with no move in government yields, credit deterioration cost 3% — that's spread risk.

Example 3 — relative value. Two BBB bonds in the same sector: Bond A at 180 bp, Bond B at 240 bp, similar duration and fundamentals. If the 60 bp gap isn't justified by any real difference (liquidity, structure), Bond B is *cheap* — you'd buy B and expect its spread to tighten toward A.

How it is tested in interviews

- **"What is a credit spread?"** — "The extra yield over the risk-free rate that compensates for default and liquidity risk. Yield = risk-free + spread."
- **"What makes spreads widen?"** — "Deteriorating fundamentals, a weakening economy, rising risk aversion, or an issuer downgrade — all raise required compensation, so spreads widen and prices fall."
- **"How does a bond's price move if its spread widens 50 bp?"** — "Down by roughly spread duration × 50 bp; for a duration of 6 that's about 3%."
- **"IG vs high yield?"** — "BBB− and above vs BB+ and below; high yield has much wider spreads, tighter covenants, and is spread/default-driven rather than rates-driven."
- **"How do you find value in credit?"** — "Compare a bond's spread to its fair spread from fundamentals and to peers/its own CDS; buy where the spread over-pays for the risk, mindful of the credit cycle."

Traps & common mistakes

- Confusing **yield** with **spread** — spread is yield *minus the risk-free rate*.
- Ignoring **spread duration** when sizing credit risk.
- Forgetting the **credit cycle** — tight spreads late-cycle are the most dangerous.
- Treating **OAS** and nominal spread as the same (OAS strips out option effects).
- Overlooking **liquidity** as a component of spread (illiquid bonds pay more).

First-principles recap

- Corporate yield = risk-free rate + credit spread; the spread pays for default + liquidity risk.
- Spreads widen (prices fall) as risk rises; credit P&L ≈ −spread duration × Δspread.
- Ratings map to spreads; the IG/HY boundary is a step-change.
- Value = spread over-paying vs fair/peers/CDS; respect the credit cycle.
- Spreads gap wider in downturns — the dominant driver of credit returns.

Quick-reference

Item	Note
Yield	Risk-free + credit spread
Spread compensates	Expected loss + risk premium + liquidity
Z-spread / OAS	Curve spread / option-adjusted
Credit P&L	≈ −spread duration × Δspread
IG vs HY	BBB− and above vs BB+ and below

Securitization & Structured Credit

The Problem / Why this matters

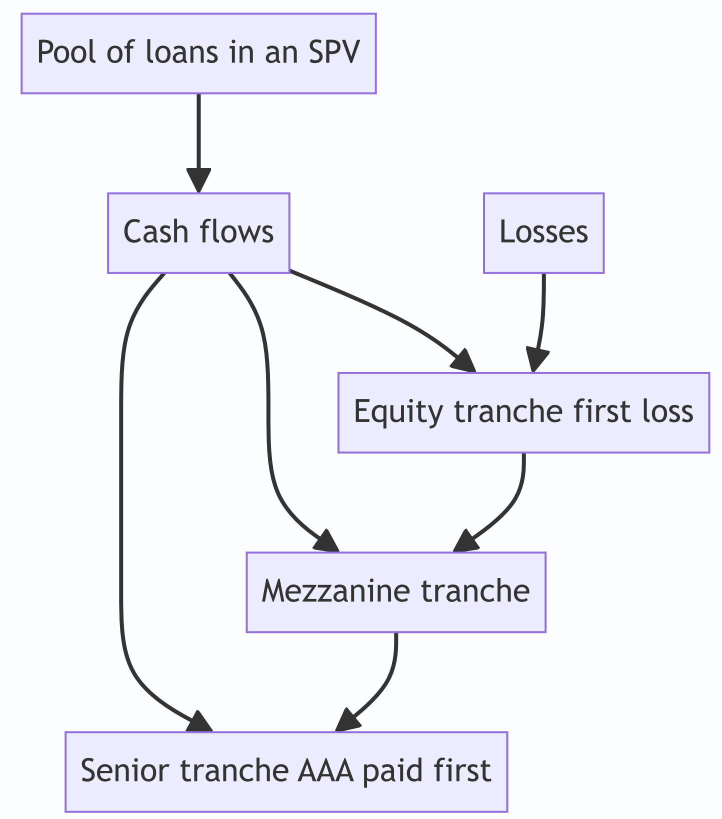
A bank with thousands of illiquid loans (mortgages, auto loans, receivables) has capital tied up and risk concentrated. Securitization lets it **pool those loans and sell them as tradable securities**, freeing capital and transferring risk to investors who want it. Done well, it channels credit efficiently and lets investors buy precisely the risk-return they want. Done badly — with opaque, mispriced, over-rated tranches — it was at the centre of the 2008 crisis. Every credit and risk interview expects you to explain tranching and the 2008 lessons.

Core Idea

Securitization pools many loans into a special-purpose vehicle (SPV) and issues securities backed by the pool's cash flows, **sliced into tranches by risk**. Senior tranches are paid first (safest, lowest yield); the equity/junior tranche absorbs first losses (riskiest, highest yield). Investors pick the tranche matching their risk appetite.

Why it works this way

Individually, loans are illiquid and each carries idiosyncratic risk. Pooling diversifies idiosyncratic risk; **tranching** then redistributes the pool's losses so that most of the structure can be made low-risk (senior) at the cost of concentrating losses in a thin equity slice. This "credit enhancement by subordination" is what lets a pool of BBB-ish loans support a large AAA-rated senior tranche.



Full technical content

The mechanics: 1. An **originator** (bank/NBFC) sells a pool of loans to a bankruptcy-remote **SPV** (true sale, off the originator's balance sheet). 2. The SPV issues **tranches** of notes backed by the pool's cash flows. 3. **The waterfall:** interest and principal flow top-down (senior first); **losses** flow bottom-up (equity/first-loss first, then mezzanine, then senior). 4. **Credit enhancement** makes the senior tranche safe: subordination (junior tranches below it), over-collateralization, excess spread, and reserve accounts.

Product zoo: | Product | Underlying pool | |---|---| | **ABS** | Auto loans, credit cards, consumer receivables | | **RMBS / MBS** | Residential mortgages | | **CMBS** | Commercial mortgages | | **CLO** | Leveraged corporate loans | | **CDO** | A pool of bonds/ABS tranches (a securitization of securitizations) |

Why tranching creates AAA from BBB. The senior tranche only takes a loss after all junior tranches are wiped out. If the equity + mezzanine tranches are (say) 20% of the structure, the pool must lose more than 20% before the senior 80% is touched — so the senior can be rated far above the average pool quality. This is powerful but relies on the **loss and correlation assumptions being right**.

The 2008 lessons (what went wrong): - **Poor underlying quality** — subprime mortgages with weak underwriting fed the pools ("garbage in"). - **Correlation underestimated** — models assumed home prices in different regions were roughly independent; in a national housing bust they fell *together*, so diversification vanished and even senior tranches took losses. - **Rating over-reliance & opacity** — investors trusted AAA labels on complex CDOs they didn't understand; CDOs-of-CDOs layered leverage on leverage. - **Misaligned incentives** — originate-to-distribute meant originators didn't keep the risk, so underwriting standards collapsed.

Post-crisis reforms: **risk retention** ("skin in the game," originators keep ~5%), better disclosure, simpler/standardized structures, and tougher rating scrutiny.

Worked examples

Example 1 — tranching and loss allocation. An SPV holds a ₹1,000 cr loan pool: senior ₹800 cr (AAA), mezzanine ₹150 cr, equity ₹50 cr. If the pool loses ₹40 cr, the **equity** tranche absorbs it all (loses 80% of its ₹50 cr); senior and mezzanine are untouched. If losses reach ₹120 cr, equity is wiped (₹50 cr) and mezzanine takes ₹70 cr (47% loss); senior still safe. Senior only bleeds if losses exceed ₹200 cr (20% of the pool).

Example 2 — how AAA is manufactured. With 20% subordination beneath it, the ₹800 cr senior tranche is protected against the first 20% of pool losses. Historically such losses were rare for the pool type, so agencies rated it AAA — *provided* the 20% cushion and the loss/correlation assumptions hold.

Example 3 — 2008 correlation failure. Models assumed regional mortgage defaults were largely uncorrelated, so a 20% cushion looked ample. When US house prices fell nationally, defaults spiked *everywhere at once* — realized correlation approached 1, losses blew through the cushion, and "AAA" tranches took losses. The lesson: **correlation, not just average default rate, drives senior-tranche risk.**

How it is tested in interviews

- **"What is securitization?"** — "Pooling illiquid loans in an SPV and issuing tradable securities backed by their cash flows, tranched by risk. It frees the originator's capital and transfers risk to investors."
- **"How does tranching work / how do you get AAA from a BBB pool?"** — "Losses hit the equity tranche first, then mezzanine, then senior. Subordination gives the senior a loss cushion, so it can be

rated well above the pool's average quality."

- **"What went wrong in 2008?"** — "Weak subprime underlying, underestimated default correlation (regional diversification failed in a national bust), over-reliance on AAA ratings for opaque CDOs, and originate-to-distribute misaligning incentives."
- **"What is a CLO?"** — "A securitization of leveraged corporate loans, tranched senior to equity."

Traps & common mistakes

- Thinking tranching **removes** risk — it **redistributes** it; total pool risk is unchanged.
- Ignoring **correlation** — the key driver of senior-tranche risk (2008's lesson).
- Trusting the **AAA label** without understanding the collateral and structure.
- Confusing **ABS/MBS** (loans) with **CDO** (a pool of bonds/tranches).
- Forgetting **originate-to-distribute** incentive problems (now mitigated by risk retention).

First-principles recap

- Securitization pools loans in an SPV and issues **tranched** securities; senior paid first, equity absorbs first losses.
- Subordination manufactures a large safe senior tranche from a riskier pool.
- Tranching redistributes, not removes, risk — total risk is conserved.
- **Correlation** drives senior-tranche risk; underestimating it caused 2008.
- Reforms: risk retention, disclosure, simpler structures.

Quick-reference

Term	Note
SPV	Bankruptcy-remote issuer of the notes
Waterfall	Cash top-down; losses bottom-up
Tranches	Senior (AAA) → mezzanine → equity (first loss)
Enhancement	Subordination, over-collateralization, excess spread
ABS/MBS/CLO/CDO	Consumer / mortgage / loans / bonds-tranches
2008 lesson	Correlation + underwriting, not just default rate

Counterparty Credit Risk & CVA

The Problem / Why this matters

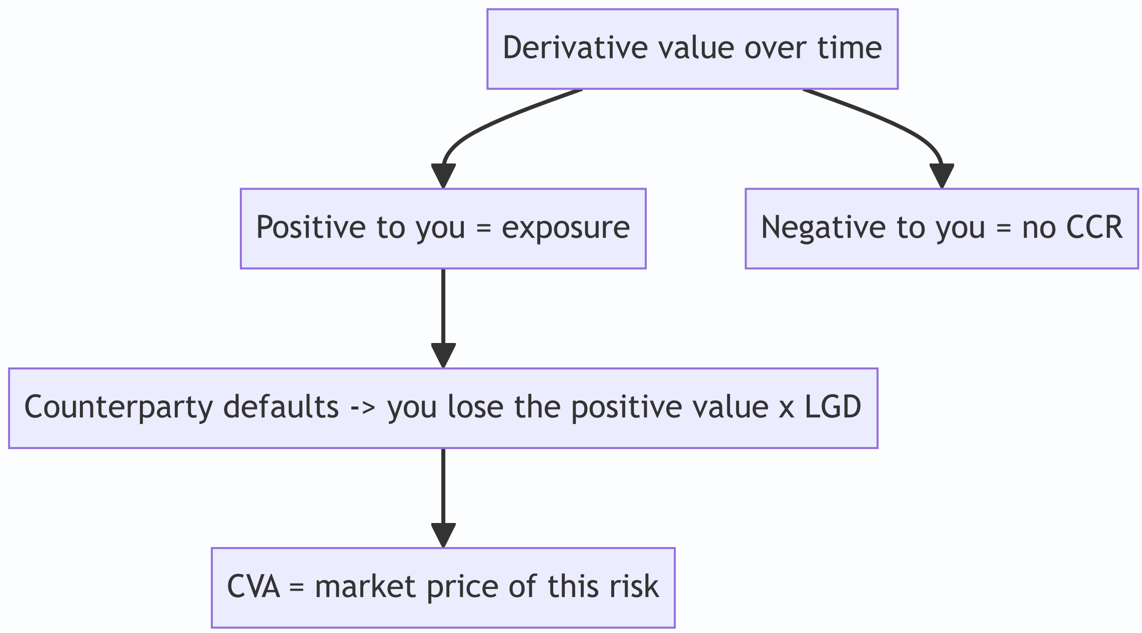
When two banks enter a 10-year interest-rate swap, neither lends the other money up front — yet each is exposed to the other. If the swap moves in your favour, the counterparty *owes* you, and if they default before paying, you lose that gain. This is **counterparty credit risk**: the risk that the other side of a derivative defaults while the trade is worth money to you. It nearly sank the financial system in 2008 (AIG, Lehman), and pricing it — via **CVA** — is now core to any derivatives, risk, or quant-credit role.

Core Idea

Counterparty credit risk (CCR) is the risk that a derivatives counterparty defaults before settling, when the contract has positive value to you. Because a derivative's value swings over its life, the exposure is uncertain and two-sided. **CVA (Credit Valuation Adjustment)** is the market price of that risk — the amount you deduct from a derivative's risk-free value to reflect the counterparty's default risk.

Why it works this way

Unlike a loan (where exposure = amount lent), a derivative's exposure **changes with the market**. Today the swap may be worth +₹5 cr to you; next year, ±₹20 cr. You only lose on default if the trade is *in your favour* at that moment — so exposure is the positive part of a random future value. Since default risk has a price (a spread), the fair value of a derivative with a risky counterparty must be *lower* than with a risk-free one; CVA is that difference.



Full technical content

Exposure measures: | Measure | Meaning | |---|---| | Current exposure | max(mark-to-market, 0) today | | **Potential Future Exposure (PFE)** | A high-percentile of possible future exposure (how bad it could get) | | Expected Exposure (EE) | Average positive exposure at a future date | | Expected Positive Exposure (EPE) | Time-average of EE (used in capital) |

CVA — the price of counterparty risk. Conceptually: **CVA ≈ Σ (discounted Expected Exposure) × (counterparty PD) × LGD** across future time buckets. It rises with (i) larger/more volatile exposure, (ii) higher counterparty default probability (wider CDS spread), and (iii) higher LGD. The **xVA** family extends this: DVA (own default), FVA (funding), MVA (margin), KVA (capital).

Risk mitigants: - **Netting** — under an ISDA master agreement, all trades with a counterparty net to a single exposure, so in-the-money and out-of-the-money trades offset (huge reduction). - **Collateral / margin (CSA)** — counterparties post collateral as the mark-to-market moves (variation margin) plus a buffer (initial margin), cutting exposure to near zero between margin calls. - **Central clearing (CCPs)** — standardized derivatives are novated to a central counterparty that margins both sides and mutualizes losses; post-2008 reform pushed most vanilla derivatives to CCPs.

Wrong-way risk. The dangerous case where exposure and counterparty default probability are **positively correlated** — e.g., buying protection on a company from a counterparty highly correlated to that company (AIG selling CDS on assets that fell exactly when AIG itself weakened). Exposure is largest precisely when the counterparty is most likely to default.

Worked examples

Example 1 — exposure is one-sided. You have a swap worth +₹8 cr to you. If the counterparty defaults now, you lose that ₹8 cr × LGD (say 60%) = ₹4.8 cr. If instead the swap were worth −₹8 cr to you (you owe them), their default costs you nothing — you'd simply settle what you owe. *Counterparty risk only bites when the trade is in your favour.*

Example 2 — netting. With one counterparty you have Trade A at +₹20 cr and Trade B at −₹15 cr. Without netting, your exposure is ₹20 cr (you ignore what you owe on B). With an ISDA netting agreement, exposure = max(20 − 15, 0) = **₹5 cr** — a 75% reduction. Netting is the single biggest CCR mitigant.

Example 3 — CVA and spread. Two identical swaps, expected exposure profile the same. Counterparty X has a CDS spread of 100 bp (low PD); counterparty Y trades at 500 bp (high PD). The CVA charge on the trade with Y is roughly 5× that with X — so you'd quote Y a worse price (or demand collateral) to compensate for the higher counterparty risk.

How it is tested in interviews

- **"What is counterparty credit risk?"** — "The risk that a derivatives counterparty defaults before settling when the trade is in your favour. Exposure is uncertain because the derivative's value moves with the market — it's the positive part of a random future value."
- **"What is CVA?"** — "The market price of counterparty default risk — the adjustment that lowers a derivative's value from its risk-free value. It's roughly expected exposure × counterparty PD × LGD."
- **"How is counterparty risk mitigated?"** — "Netting under an ISDA master, collateral/margin under a CSA, and central clearing through CCPs."

- **"What is wrong-way risk?"** — "When exposure rises exactly as the counterparty's default probability rises — they're positively correlated, like AIG selling protection on assets tied to its own health."

Traps & common mistakes

- Treating derivative exposure like a **loan** (fixed) rather than a **random, market-driven** amount.
- Forgetting exposure is **one-sided** (only positive MtM matters).
- Ignoring **netting and collateral** when sizing exposure.
- Overlooking **wrong-way risk** — the most dangerous CCR.
- Confusing **CVA** (counterparty's default) with **DVA** (your own).

First-principles recap

- CCR = a derivatives counterparty defaults while the trade is in your favour.
- Exposure is the **positive** part of an uncertain future value (PFE/EE).
- **CVA** prices it ≈ expected exposure × PD × LGD; it widens with the counterparty's CDS spread.
- Mitigated by **netting, collateral/margin, and central clearing**.
- **Wrong-way risk** (exposure correlated with default) is the tail danger.

Quick-reference

Term	Note
CCR	Counterparty defaults with trade in your favour
Exposure	max(MtM, 0); PFE = high-percentile future
CVA	≈ EE × PD × LGD (price of CCR)
Mitigants	Netting (ISDA), collateral (CSA), CCP clearing
Wrong-way risk	Exposure ↑ as counterparty PD ↑

Portfolio Credit Risk & Concentration

The Problem / Why this matters

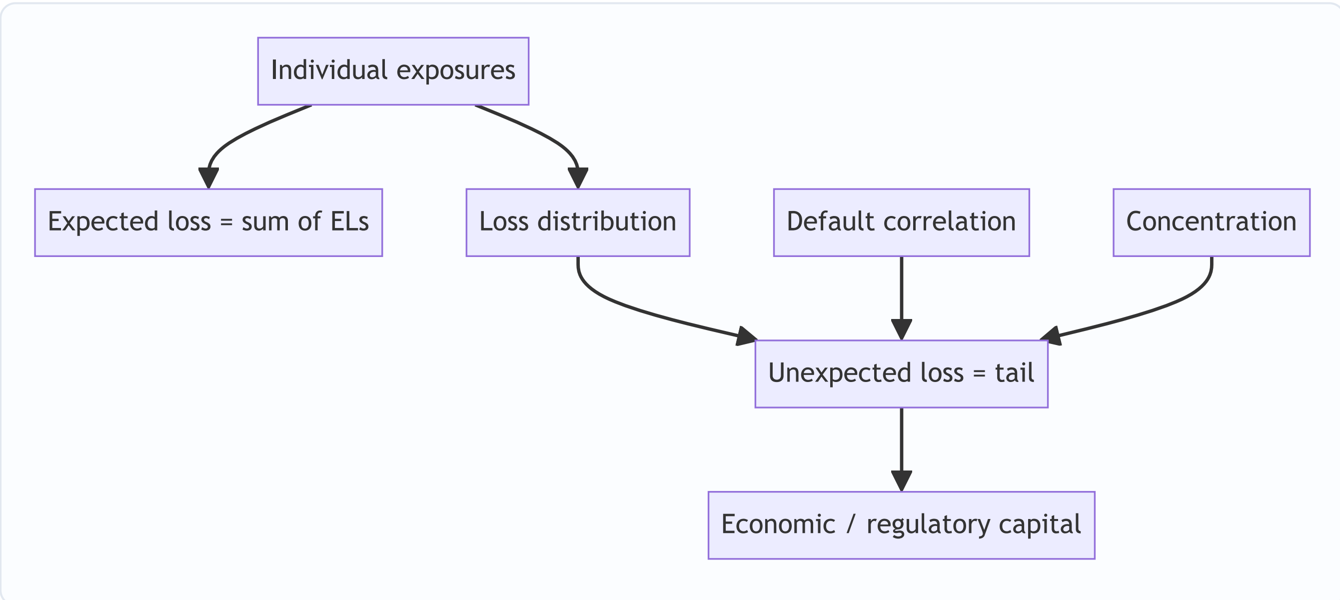
A single loan can be analysed borrower-by-borrower, but a bank holds thousands. What matters to the *institution* is not any one default but the **loss distribution of the whole portfolio** — and, crucially, whether losses arrive independently or all at once. Two portfolios with identical average default rates can have wildly different risk if one is concentrated and correlated. Managing **correlation and concentration** is what separates a bank that survives a downturn from one that doesn't. This is core to portfolio-management, risk, and NBFC roles.

Core Idea

Portfolio credit risk is the loss distribution of a book of exposures. **Expected loss** is just the sum of individual expected losses, but **unexpected loss** — the tail — is driven by **default correlation** and **concentration**. Diversification reduces unexpected loss; correlation and concentration amplify it. Capital is held against the tail, not the average.

Why it works this way

If defaults were independent, a large portfolio's loss would be very stable (the law of large numbers) and you'd need little capital. But borrowers share common risk factors — the economy, a sector, a region — so they tend to default *together* in downturns. That common factor creates correlation, fattens the tail of the loss distribution, and is the reason banks need substantial capital and stress testing.



Full technical content

Expected vs unexpected portfolio loss: - **Portfolio EL** = Σ ($PD_i \times LGD_i \times EAD_i$) — additive, unaffected by correlation. - **Portfolio UL** — the standard deviation / tail of the loss distribution — **depends heavily on correlation**. Higher correlation → fatter tail → more capital.

Credit VaR / economic capital. Just as market VaR, **Credit VaR** is a high-percentile (e.g., 99.9%) of the portfolio loss distribution over a horizon. **Economic capital ≈ Credit VaR − EL** (capital covers *unexpected* loss beyond the expected, which pricing/provisions already cover). Basel's IRB formula is a single-factor version of exactly this.

What drives the tail: | Driver | Effect | |---|---| | **Default correlation** | Higher correlation → losses cluster → fatter tail | | **Name concentration** | A few large exposures → single defaults move the whole book | | **Sector concentration** | Many borrowers in one industry default together | | **Geographic concentration** | Regional shock hits many at once | | **Systematic factor exposure** | Sensitivity to the common economic factor |

Diversification and its limits. Spreading across uncorrelated names, sectors and regions reduces UL — but you can only diversify away *idiosyncratic* risk. The **systematic** component (common exposure to the economy) cannot be diversified away, which is why even a broad book takes heavy losses in a severe recession.

Managing concentration: single-name limits, sector/geography caps, correlation-aware limits, and portfolio actions (loan sales, securitization, credit default swaps, index hedges) to shed concentrated risk.

Stress testing the book. Because tail losses come from common factors, banks stress the whole portfolio against scenarios (a recession, a sector collapse, a rate shock) to see clustered losses that normal VaR under models — regulators require it.

Worked examples

Example 1 — correlation fattens the tail. Two 100-loan portfolios, each loan PD 2%, LGD 50%, EAD ₹1 cr. Both have **EL = 100 × 0.02 × 0.5 × 1 = ₹1 cr**. Portfolio A's defaults are near-independent → in a bad year maybe 4–5 defaults (loss ~₹2–2.5 cr). Portfolio B's borrowers all serve one cyclical sector (high correlation) → in a downturn 20 default *together* (loss ~₹10 cr). **Same expected loss, 4× the tail** — B needs far more capital.

Example 2 — concentration. A ₹1,000 cr book: Portfolio X has 100 loans of ₹10 cr each; Portfolio Y has 10 loans of ₹100 cr each. Same total. One default in X costs ₹10 cr (1% of book); one default in Y costs ₹100 cr (10%). Y's lumpiness makes single defaults material — concentration risk, even at the same average PD.

Example 3 — diversification limit. A well-diversified retail book has low idiosyncratic risk, but every borrower is exposed to the domestic economy. In a national recession, defaults rise across the whole book simultaneously — the *systematic* loss that no amount of name diversification removes. This is why the tail, not the average, sizes capital.

How it is tested in interviews

- **"How is portfolio credit risk different from single-name?"** — "Expected loss is just the sum of individual ELs, but the risk that matters — the tail / unexpected loss — is driven by default correlation and concentration, not the average."
- **"Why does correlation matter so much?"** — "It determines whether defaults arrive independently or cluster. High correlation fattens the loss tail, so you need much more capital for the same expected loss."
- **"Two books, same average PD — which is riskier?"** — "The more concentrated and correlated one — a few large, sector-clustered exposures move the whole portfolio together."

- **"What can you diversify away in a credit portfolio?"** — "Idiosyncratic, name-specific risk. The systematic exposure to the economy can't be diversified away — which is why severe recessions hit even broad books."
- **"What is economic capital for credit?"** — "Roughly Credit VaR minus expected loss — capital held against unexpected (tail) losses."

Traps & common mistakes

- Assuming defaults are **independent** — they cluster via common factors.
- Sizing risk on **average** PD and ignoring the **tail**.
- Missing **concentration** (name, sector, geography).
- Believing diversification removes **all** risk — the systematic part remains.
- Using normal-market **VaR** without **stress testing** for clustered downturn losses.

First-principles recap

- Portfolio EL is additive; **UL (the tail) is driven by correlation and concentration**.
- Capital covers unexpected loss (Credit VaR − EL), not the average.
- Diversification removes idiosyncratic, not systematic, risk.
- Concentration (name/sector/geography) makes single events move the whole book.
- Stress test for clustered, downturn losses that models understate.

Quick-reference

Concept	Note
Portfolio EL	$\Sigma\ PD_i \times LGD_i \times EAD_i$
Unexpected loss	Tail; driven by correlation
Credit VaR	High-percentile portfolio loss
Economic capital	≈ Credit VaR − EL
Diversifiable	Idiosyncratic only (not systematic)
Controls	Limits, caps, loan sales, CDS, securitization

A Credit Case: Building an Internal Rating & Credit Memo (Capstone)

The Problem / Why this matters

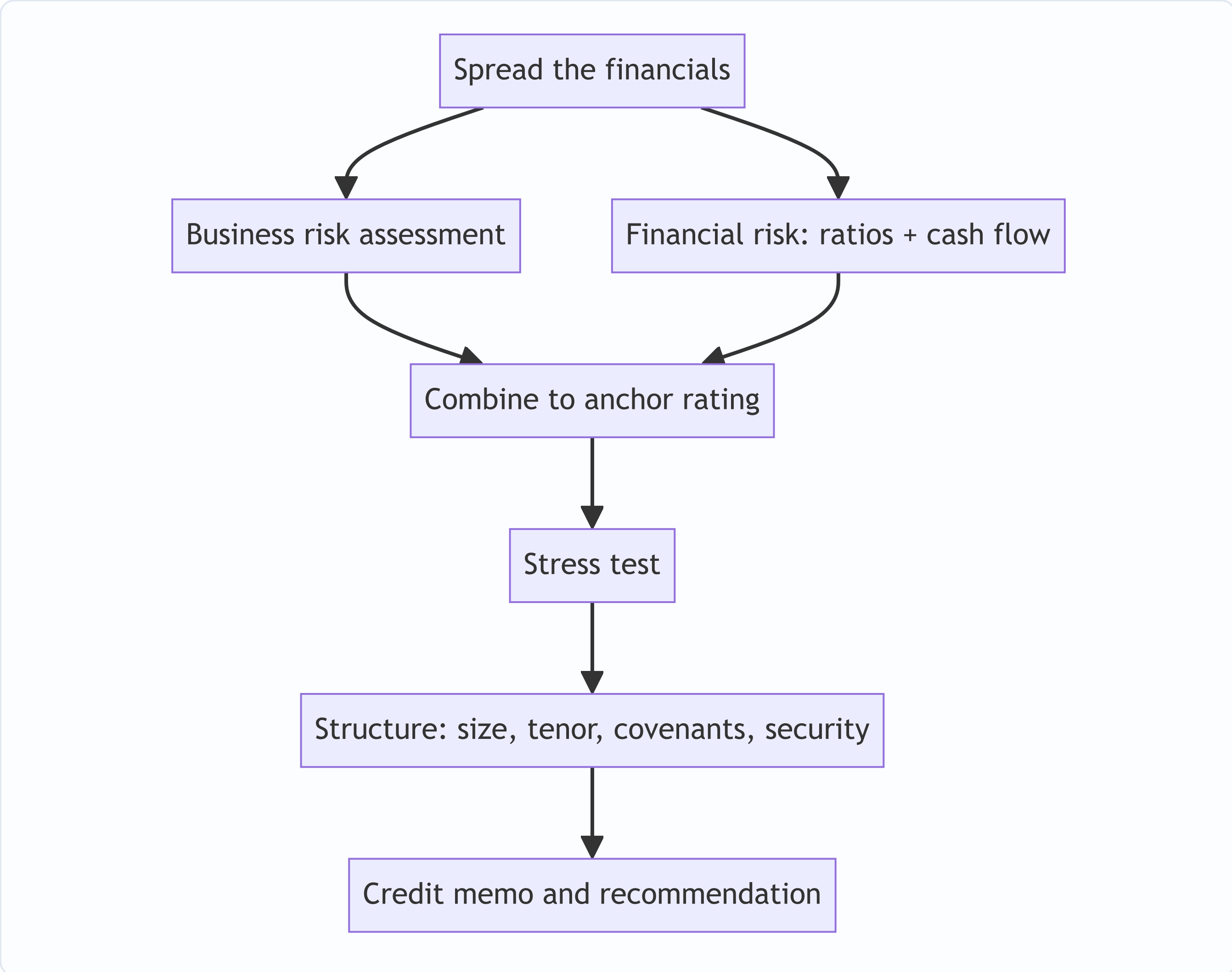
Everything in this book — spreading, ratios, cash flow, business risk, structure, PD/LGD, recovery — comes together in one deliverable: the **credit memo** that recommends whether and how to lend, and the **internal rating** that anchors the pricing and the capital. This capstone walks a complete case end to end, exactly as you'd do it on the job and describe it in an interview when asked "how would you assess this borrower?"

Core Idea

A credit assessment flows in a fixed sequence: **spread → analyse (business + financial risk) → stress → rate → structure → recommend**. The internal rating combines a business-risk score and a financial-risk score into an anchor grade; the memo documents the analysis, the rating, the proposed structure, the risks, and the recommendation.

Why it works this way

Credit committees need a consistent, auditable path from raw financials to a decision. Structuring the analysis the same way every time forces completeness (nothing skipped), comparability (this borrower vs the book), and accountability (the rationale is written down). The rating makes risk comparable and drives pricing and capital.



Full technical content — the worked case

The borrower: MidCo, an auto-components manufacturer seeking a ₹150 cr term loan for capacity expansion.

Step 1 — Spread & normalize. Revenue ₹800 cr; reported EBITDA ₹130 cr, but it includes a ₹15 cr one-off land-sale gain and excludes ₹5 cr of operating-lease rent → **normalized EBITDA ₹120 cr**. Reported debt ₹300 cr; add ₹40 cr factored receivables and ₹20 cr capitalized leases → **adjusted debt ₹360 cr**. Tangible net worth ₹200 cr.

Step 2 — Financial risk (ratios & cash flow). - Leverage: adjusted Debt/EBITDA = 360/120 = **3.0x**; D/E = 360/200 = **1.8x**. - Coverage: EBIT (₹95 cr) / interest (₹30 cr) = **3.2x**. - Cash flow: EBITDA 120 − cash tax 20 − ΔWC 15 − maintenance capex 20 = **CADS ₹65 cr**; DSCR = 65 / (interest 30 + principal 20) = 65/50 = **1.3x**. - Read: moderate leverage (3.0x), adequate coverage (DSCR 1.3x), positive CADS. *Financial risk: intermediate.*

Step 3 — Business risk. Auto components: **cyclical** (tied to auto demand), moderately competitive, but MidCo is a **top-3 supplier** to two large OEMs with long-standing relationships. Risks: **customer concentration** (60% from two OEMs), commodity (steel) cost pass-through lag, EV-transition risk to legacy parts. *Business risk: intermediate-to-high, chiefly from cyclicality and concentration.*

Step 4 — Anchor rating. Intermediate business risk × intermediate financial risk → an anchor around **BBB (internal grade ~4 on a 1–8 scale)**. Modifiers: adequate liquidity (+), experienced management with a clean track record (+), customer concentration (−). Net internal rating ≈ **BBB / grade 4**.

Step 5 — Stress test. Downside: auto demand falls, EBITDA drops 25% to ₹90 cr; steel costs squeeze margins. Stressed CADS ≈ ₹45 cr; stressed DSCR = 45/50 = **0.9x** — a shortfall. *Conclusion:* the base case is serviceable but a moderate downturn breaches DSCR — so **structure must build in cushion and controls.**

Step 6 — Structure the facility. - **Size** ₹150 cr (post-facility Debt/EBITDA ≈ (360+150)/? — actually expansion lifts EBITDA; size to keep stressed DSCR near 1.2x). Right-size and/or phase drawdown. - **Tenor** 7 years, matching the asset life; **amortization** front-loaded/straight-line while cash flow is strong (avoid a bullet given cyclicality). - **Security** first charge on the new plant + hypothecation of current assets; corporate guarantee. - **Covenants** (maintenance): Debt/EBITDA ≤ 3.5x (stepping down), DSCR ≥ 1.2x, minimum net worth, capex cap, and a restricted-payments limit; cross-default. - **Pricing** spread reflecting the BBB internal rating and expected loss.

Step 7 — The credit memo (structure of the write-up): 1. Executive summary & recommendation (approve/decline, amount, rating). 2. Transaction & purpose. 3. Borrower & management overview. 4. Business & industry risk. 5. Financial analysis (spread, ratios, cash flow, trend). 6. Stress test & sensitivities. 7. Proposed structure, security & covenants. 8. Internal rating & pricing. 9. Key risks & mitigants. 10. Recommendation.

The recommendation: *Approve ₹150 cr, internal rating BBB/grade 4, 7-year amortizing, first charge + guarantee, maintenance covenants (Debt/EBITDA ≤ 3.5x, DSCR ≥ 1.2x), priced at [spread]. Key risks: cyclicality and customer concentration, mitigated by front-loaded amortization, covenants, security, and phased drawdown.*

How it is tested in interviews

- **"How would you assess a borrower / walk me through a credit."** — Recite the sequence: **spread & normalize → financial risk (ratios + cash flow) → business risk → anchor rating → stress test → structure & covenants → recommendation.** Then apply it to whatever example they give.
- **"How do you decide how much to lend?"** — From stressed cash flow and a target DSCR, sized to keep stressed DSCR near 1.2x, cross-checked against a leverage cap.
- **"This borrower has 3x leverage and 1.3x DSCR — approve?"** — "Base case is serviceable, but I'd stress it: if a moderate downturn takes DSCR below 1.0x, I approve only with structure — front-loaded amortization, tight covenants, security, and possibly a smaller/phased facility."
- **"What goes in a credit memo?"** — The ten sections above, ending in a clear rating and recommendation.

Traps & common mistakes

- Jumping to a decision **without the stress test** — the base case always looks fine.
- Rating on **financials alone**, ignoring business risk (cyclicality, concentration).
- Sizing the loan off **base-case** rather than **stressed** cash flow.
- A memo that **describes** without **recommending** — the committee needs a clear call and rating.
- Forgetting to tie **structure and covenants** back to the specific risks identified.

First-principles recap

- The credit flow: **spread → analyse (business + financial) → rate → stress → structure → recommend.**
- The internal rating combines business-risk and financial-risk scores into an anchor, adjusted by modifiers.
- Size and structure off the **stressed** case to hold DSCR above ~1.2x.
- Tie every covenant and security feature to a specific identified risk.
- The memo ends in a clear rating and a clear recommendation.

Quick-reference — the credit checklist

Step	Output
Spread & normalize	Clean recurring numbers
Financial risk	Leverage, coverage, DSCR, CADS
Business risk	Industry + position + concentration
Anchor rating	Business × financial → grade
Stress test	Stressed DSCR/leverage
Structure	Size, tenor, amortization, security, covenants
Memo	10 sections → rating + recommendation